

AUTOMATION AND RENT DISSIPATION: IMPLICATIONS FOR WAGES, INEQUALITY, AND PRODUCTIVITY*

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This article studies the effects of automation in a task-based economy in which some jobs pay workers rents—wages above workers’ outside options. We show that automation targets high-rent tasks, dissipating rents, amplifying wage losses, and reducing within-group wage dispersion in exposed groups. This form of rent dissipation is inefficient and offsets the productivity gains from automation. Using U.S. data from 1980 to 2016, we find evidence of sizable rent dissipation and reduced within-group wage dispersion due to automation. Automation accounts for 52% of the increase in between-group inequality since 1980, with rent dissipation explaining one-fifth of this total. Our estimates imply that inefficient rent dissipation has offset 60%–90% of the productivity gains from automation over this period. *JEL codes:* J23, J31, O33.

I. INTRODUCTION

The U.S. labor market has undergone significant changes since 1980: not only has inequality increased sharply, but real wages for workers without a college degree have stagnated or declined. Although many factors may have contributed to rising inequality, the automation of tasks performed by less educated workers appears to have played an important role.¹

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1. See Goldin and Katz (2008), Acemoglu and Autor (2011), and Autor (2019) for a review of U.S. inequality trends, and Acemoglu and Restrepo (2022b) for estimates of the impact of automation on U.S. inequality.

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This article examines the implications of automation for wages, productivity, and welfare in economies with imperfect labor markets, where workers earn rents—wages above their outside options or opportunity costs. While the presence of worker rents is well-documented, the literature has not reached consensus on how changes in these rents have shaped the wage structure, and little is known about how the interplay between automation and labor market imperfections has affected inequality.² Our contribution is to develop a new framework for analyzing, estimating, and quantifying the effects of automation in the presence of worker rents.

We consider an economy where firms allocate tasks to workers of different skills or automate them. Worker rents distort hiring and automation decisions by creating a wedge between the wage firms must pay workers in some tasks and their opportunity cost. This wedge may be due to efficiency wages, bargaining, regulations (such as licensing requirements), or norms. These wedges artificially inflate the cost of hiring workers, reducing employment in high-rent tasks below the efficient level and encouraging excessive automation of these tasks.³

Our first theoretical contribution is to identify a novel rent dissipation mechanism. All else equal, automation targets and displaces workers from high-rent tasks. This targeting has important implications for wages and productivity:

- (i) *Average group wages.* In competitive labor markets, automation depresses the relative demand for exposed workers via a displacement effect—by reducing the share of tasks allocated to them. Rent dissipation amplifies wage losses by pushing exposed groups away from high-rent tasks toward lower-wage jobs.
- (ii) *Within-group wage dispersion.* By reallocating workers away from high-rent jobs, automation reduces within-

2. The presence of worker rents receives support from the literature on wage losses following job displacement (Gibbons and Katz 1992; Jacobson, LaLonde, and Sullivan 1993; Schmieder, von Wachter, and Heining 2023), earnings differentials (Krueger and Summers 1988; Katz and Summers 1989; Card et al. 2018), and wage premia associated with unions and licenses (Kleiner and Krueger 2013; Gittleman and Kleiner 2016; Farber et al. 2021).

3. This definition of rents excludes compensating differentials, paid to compensate workers for amenities or hazards, and high wages accruing to workers with specialized skills.

group wage dispersion. This results in a U-shaped profile of wage changes across percentiles of exposed groups.

- (iii) *Productivity and welfare.* In competitive labor markets, automation increases total factor productivity (TFP) by reducing the cost of producing automated tasks. In our economy, rent dissipation is inefficient. Tasks targeted for automation are those where worker wages are high due to rents, not due to scarcity or skills. A planner would have preferred to allocate more—rather than less—labor to these tasks. Inefficient rent dissipation offsets some (or all) of the cost-saving gains from automation.

Our second contribution is to provide reduced-form evidence on rent dissipation due to automation using U.S. data from 1980 to 2016. In a first strategy, we document that the negative effect of automation on wages within detailed demographic groups follows the distinctive U-shaped pattern predicted by theory, with the largest effects between the 70th and 95th percentiles of the within-group distribution. This is indicative of automation leading to a loss of high-rent jobs.

In a second strategy, we directly show that automation displaced exposed groups of workers from high-rent jobs, using various proxies of rents proposed in the literature (see [Krueger and Summers 1988](#); [Katz and Summers 1989](#); [Gibbons and Katz 1992](#)). These include wage differentials across industries and occupations (controlling for worker characteristics), wage losses after job displacement, and quit rates (an inverse measure of rents, since workers leave attractive jobs less often). Using these proxies, we document that the U.S. workforce has shifted over time from high-rent jobs toward lower-rent jobs, with the shift more pronounced for groups exposed to automation.

Both strategies yield robust results, even after controlling for compensating wage differentials and other factors, such as minimum wages. They also imply a sizable role for rent dissipation: on average, automation displaced workers from jobs where wages exceeded their opportunity cost by about 35%. This loss of rents accounts for one-fifth of the overall relative wage decline experienced by groups exposed to automation.

Our third contribution is to quantify the general equilibrium implications of automation for wages and productivity in the presence of labor market distortions. We derive formulas that express the effects of automation in terms of task displacement (the share

of tasks automated for a demographic group), the extent of rent dissipation, and cost savings from automation. In general equilibrium, the automation of tasks performed by a group affects other groups via ripple effects—as the displaced group competes against others for marginal tasks more intensively. Our formulas summarize these ripple effects using two matrices, which can be estimated using U.S. data: the propagation matrix, which encodes this competition for marginal tasks, and the rent-impact matrix, which encodes information on how task reallocation changes rents across groups.

We find that automation accounts for 52% of the rise in between-group wage inequality since 1980. Forty-two percentage points of the 52 are due to the baseline effects of automation working through labor demand. The remaining 10 percentage points are due to rent dissipation. We also estimate that the cost savings from automation increased TFP by 3% between 1980 and 2016. However, the inefficient targeting of high-rent tasks offsets 60%–90% of these gains. On net, automation increased aggregate TFP by only 0.3%–1.3% and aggregate consumption by just 0.45%–1.95% during this period.

I.A. Literature

Our main contribution is to develop a framework for analyzing the effects of automation on labor markets with rents. Several papers study the interplay between technology and labor market imperfections (e.g., [Aghion and Howitt 1994](#); [Acemoglu 1997](#); [Caballero and Hammour 1998](#); [Mortensen and Pissarides 1998](#)), but they do not explore the effects of automation. Recent exceptions include [Arnoud \(2019\)](#) and [Leduc and Liu \(2022\)](#), who focus on how automation affects wages via bargaining.

Our framework extends the task model in [Acemoglu and Restrepo \(2022b\)](#) to incorporate worker rents.⁴ We model rents as exogenous wedges, following the misallocation literature (e.g., [Restuccia and Rogerson 2008](#); [Hsieh and Klenow 2009](#); [Hsieh et al. 2019](#)). This choice preserves comparability with prior analyses in competitive markets and isolates the differences due to

4. These models build on prior works exploring the effects of technologies and trade in task models, including [Zeira \(1998\)](#), [Acemoglu and Zilibotti \(2001\)](#), [Autor, Levy, and Murnane \(2003\)](#), [Acemoglu and Autor \(2011\)](#), and [Acemoglu and Restrepo \(2018\)](#), as well as [Grossman and Rossi-Hansberg's \(2008\)](#) model of offshoring.

rents. We also provide formulas for the effects of automation on wages and productivity in distorted economies, which relate to work by [Baqaee and Farhi \(2020\)](#), [Basu et al. \(2022\)](#), and [Dávila and Schaab \(2023\)](#).

Our work contributes to the literature on the rise in U.S. wage inequality (e.g., [Katz and Murphy 1992](#); [Card and Lemieux 2001](#)), particularly to papers examining the impact of computers and automation on inequality, including [Krueger \(1993\)](#); [Autor, Katz, and Krueger \(1998\)](#); [Autor, Levy, and Murnane \(2003\)](#), to the literature on capital-skill complementarity (e.g., [Krusell et al. 2000](#); [Burstein, Morales, and Vogel 2019](#)), and to our previous work [Acemoglu and Restrepo \(2022b\)](#), which modeled and quantified the effects of automation on the U.S. wage structure. In relation to these publications, our contribution is to examine the interplay between automation and worker rents.

Our emphasis on technology and rents is closer to work by [Bound and Johnson \(1992\)](#) and [Borjas and Ramey \(1995\)](#). [Bound and Johnson \(1992\)](#) decompose changes in the wage structure between 1979 and 1988 into technology, supply, and rents (modeled as exogenous industry wedges), finding that 10%–20% of between-group wage changes reflect the loss of rents. [Borjas and Ramey \(1995\)](#) estimate that between 1976 and 1990, trade increased the college premium by 1.3–2.6 log points, with 15%–33% of the effect due to loss of rents from trade-exposed jobs. Both estimates are comparable to ours, though [Bound and Johnson](#)'s estimates include all factors affecting rents. Recent work by [Stansbury and Summers \(2020\)](#) also documents a decline in worker rents but does not explore the role of automation.

We also contribute to this literature by documenting the implications of automation for between- and within-group inequality. We find that groups exposed to automation have experienced lower wage growth on average and more within-group wage compression—a phenomenon explained by a shift away from high-rent jobs. This pattern contrasts with the standard view that inequality is fractal, rising at all levels of aggregation (e.g., [Katz 1994](#)), due to higher skill demand (see [Acemoglu 2002](#)). Our theory points to a more nuanced pattern: automation can increase wage inequality between groups while reducing it in affected groups, especially among noncollege workers who have been more exposed to automation. This mechanism helps explain the divergent trends in within-group wage inequality, which has steadily risen for college workers since the 1980s while remaining flat and then declin-

ing for noncollege workers since the 1990s (see [Goldin and Katz 2007](#)).

Our exploration of the effect of technology on within-group inequality relates to recent work by [Kogan et al. \(2021\)](#) and [Danieli \(2024\)](#). These papers have little overlap with our approach, as they do not model or estimate the effects of automation via rents.

Finally, our work contributes to the literature on worker rents and their implications for efficiency. This literature proposes several reasons workers earn rents, ranging from efficiency wages ([Akerlof 1984](#); [Shapiro and Stiglitz 1984](#); [Stiglitz 1985](#); [Bulow and Summers 1986](#)) to search, bargaining ([Grout 1984](#); [Pissarides 2000](#)), and labor market regulations. This literature shows that worker rents can be distortionary, leading to insufficient employment in high-rent jobs. These distortions can justify second-best policy interventions and imply that technology or trade can reduce welfare (see [Bhagwati 1968](#); [Katz and Summers 1989](#)), as emphasized here.

I.B. Organization

[Section II](#) presents our theoretical framework. [Section III](#) documents reduced-form evidence of rent dissipation. [Section IV](#) outlines our approach for estimating the general equilibrium effects of automation. The [Online Appendix](#) includes the proofs, robustness checks, and data details.

II. THEORY: AUTOMATION AND LABOR MARKET DISTORTIONS

This section presents our framework and derives theoretical results. We study a one-sector model and extend to a multi-sector one in [Section IV](#).

II.A. Single-Sector Model

1. Description. A unique final good y is produced by combining complementary tasks $x \in \mathcal{T}$ (where the set of tasks $\mathcal{T} \subset \mathbb{R}^d$ has mass M). Task quantities, y_x , are aggregated with a constant elasticity of substitution $\lambda \in (0, 1)$.⁵

$$y = \left(\frac{1}{M} \int_{\mathcal{T}} (M y_x)^{\frac{\lambda-1}{\lambda}} dx \right)^{\frac{\lambda}{\lambda-1}}.$$

5. The set $\mathcal{T}$ is assumed measurable and $\int f_x dx$ denotes the Lebesgue integral of f_x in $\mathbb{R}^d$.

There is a finite set of labor groups (demographic groups in our application) indexed by g , where $g \in \mathbb{G} = \{1, 2, \dots, G\}$. Tasks can also be produced using task-specific capital, equipment, or software, denoted by k_x for task x . The total production of task x is

$$y_x = \psi_{kx} k_x + \sum_g \psi_{gx} \ell_{gx},$$

where ℓ_{gx} is the amount of labor of type g allocated to task x ; k_x is the amount of task-specific capital used for this task; and ψ_{gx} and ψ_{kx} represent their productivity in task x . In our baseline model, workers in a group share the same productivity profile across tasks, but differ from other groups in their comparative advantage.

There is an inelastic supply ℓ_g of workers of type g to be allocated across tasks, so that

$$\int_{\mathcal{T}} \ell_{gx} dx \leq \ell_g.$$

We treat task-specific machines, $\{k_x\}_{x \in \mathcal{T}}$, as intermediate goods. They are produced in the same period using the final good at the constant unit cost $\frac{1}{q_x}$. This implies that total consumption equals net output:

$$c = y - \int_{\mathcal{T}} \frac{k_x}{q_x} dx.$$

If $q_x = 0$, task x cannot be automated (or produced by capital).

2. Labor Market Distortions and Equilibrium. We model labor market distortions using task-specific wedges. A firm that employs labor of type g to perform task x pays a wage

$$w_{gx} = w_g \mu_{gx},$$

where $w_g > 0$ is the base wage of group g , and $\mu_{gx} \geq 1$ is an exogenous wedge that varies across tasks. The wedge μ_{gx} captures worker rents: how much more task x pays workers from group g relative to their base wage. We assume $\mu_{gx} = 1$ for some of the tasks assigned to group g , so that base wages are equal to what workers earn in tasks that pay no rents.

The wedges $\{\mu_{gx}\}$ represent distortions that force firms to pay wages above what workers could earn in other tasks. For example, μ_{gx} may capture rents from efficiency wages or bargaining, as

shown in [Online Appendix S3](#). The wedges may also capture constraints imposed by national union contracts, licensing boards, or regulations.⁶

We treat labor wedges as fixed, a convenient simplification that implies they do not adjust in response to changes elsewhere in equilibrium. Our results only require that wedges are rigid and do not fully adjust to restore efficiency—a common feature of many sources of distortions. For example, in the micro-foundations given in the [Online Appendix](#), wedges depend on the monitoring technology, bargaining parameters, or irreversible investments and do not adjust in response to automation or broader market conditions.⁷

The labor market operates as follows. Firms take base wages w_g and wedges μ_{gx} as given and decide how many g workers to hire for task x at a wage w_{gx} . Workers prefer higher-rent jobs, but these are rationed in equilibrium: firms hire until the value of the marginal product of labor (VMPL) equals the wage w_{gx} . Workers are then randomly assigned to tasks until labor demand is met, and the base wage w_g adjusts to ensure full employment.

Formally, a market equilibrium is given by base wages $w = \{w_g\}_{g \in \mathbb{G}}$, output (real GDP) y , an allocation of tasks $\{\mathcal{T}_g\}_{g \in \mathbb{G}}$, $\mathcal{T}_k$, task prices $\{p_x\}_{x \in \mathcal{T}}$, hiring plans $\{\ell_{gx}\}_{gx \in \mathbb{G} \times \mathcal{T}}$, and capital $\{k_x\}_{x \in \mathcal{T}}$ such that:

E1 Task prices equal the minimum unit cost of performing them:

$$p_x = \min \left\{ \frac{1}{q_x \psi_{kx}}, \left\{ \frac{w_g \mu_{gx}}{\psi_{gx}} \right\}_g \right\}.$$

E2 The task allocation minimizes costs. Workers from group g perform tasks in

$$\mathcal{T}_g = \left\{ x : p_x = \frac{w_g \mu_{gx}}{\psi_{gx}} \right\}$$

6. Monopsony power has implications similar to our wedges. This is because monopsony acts like a tax on firm hiring (the cost of hiring new workers exceeds the wage because this raises wages for existing workers). As a result, monopsony distortions reduce employment and create incentives for excessive automation.

7. An exception is efficient bargaining models, where workers and firms are on their contract curve, ensuring an efficient level of employment.

for every $g \in \mathbb{G}$, and capital performs the remaining tasks in

$$\mathcal{T}_k = \left\{ x : p_x = \frac{1}{q_x \psi_{kx}} \right\}.$$

- E3 Workers in g are rationed across tasks in $\mathcal{T}_g$. Firms hire ℓ_{gx} for task x , at the wage $w_{gx} = w_g \mu_{gx}$. The base wage adjusts to ensure the labor market clears $\int_{\mathcal{T}_g} \ell_{gx} dx = \ell_g$.
- E4 For every $x \in \mathcal{T}_k$, firms use capital until its marginal product value equals its cost.
- E5 The price of the final good is normalized to one, which requires:

$$1 = \left(\frac{1}{M} \int_{\mathcal{T}} p_x^{1-\lambda} dx \right)^{\frac{1}{1-\lambda}}.$$

To ensure the existence of a unique equilibrium, we impose the following assumption.

ASSUMPTION 1 (RESTRICTIONS ON THE TASK SPACE).

- For each task $x \in \mathcal{T}$, there exists at least one $g \in \mathbb{G}$ such that $\psi_{gx} > 0$.
- For each $g \in \mathbb{G}$, the set $\{x \in \mathcal{T} \text{ st: } \psi_{gx} > 0, \psi_{g'x} = 0 \text{ for all } g' \neq g \text{ and } \psi_{kx} = 0 \text{ or } q_x = 0\}$ has positive measure.
- For each $g \in \mathbb{G}$, $\int_{\{x: \psi_{gx} > 0\}} \psi_{gx}^{\lambda-1} \mu_{gx}^{-\lambda} dx$ and $\int_{\{x: \psi_{gx} > 0\}} \psi_{gx}^{\lambda-1} \mu_{gx}^{1-\lambda} dx$ are bounded.
- Comparative advantage is strict: for any two groups $g \neq g'$ and constants $a, b > 0$, the set of tasks for which $\frac{\psi_{gx}}{\mu_{gx}} = a \frac{\psi_{g'x}}{\mu_{g'x}}$ and $\frac{\psi_{gx}}{\mu_{gx}} = b q_x \psi_{kx}$ has measure zero.

Strict comparative advantage ensures that labor demand is smooth and groups are imperfect substitutes in aggregate. Combined with the other conditions, this guarantees the existence of a unique equilibrium where each group g performs a positive share of tasks.

II.B. Equilibrium with Labor Market Rents

Following [Acemoglu and Restrepo \(2022b\)](#), we characterize the equilibrium in terms of task shares. Define the task shares

of worker group g and capital as

$$\Gamma_g(w) = \frac{1}{M} \int_{\mathcal{T}_g(w)} \psi_{gx}^{\lambda-1} \mu_{gx}^{-\lambda} dx \text{ for } g \in \mathbb{G},$$

$$\Gamma_k(w) = \frac{1}{M} \int_{\mathcal{T}_k(w)} (\psi_{kx} q_x)^{\lambda-1} dx.$$

The integrals are computed over the set of tasks allocated to worker groups and capital, denoted by $\mathcal{T}_g(w)$ and $\mathcal{T}_k(w)$, when base wages are w . Task shares summarize how the range of tasks assigned to workers and capital varies as a function of base wages. [Assumption 1](#) ensures task shares are bounded, positive, and differentiable.

In addition, define the average group rent earned by g workers in tasks in $\mathcal{T}_g(w)$ as

$$\mu_g(w) = \frac{1}{M \Gamma_g(w)} \int_{\mathcal{T}_g(w)} \psi_{gx}^{\lambda-1} \mu_{gx}^{1-\lambda} dx \text{ for } g \in \mathbb{G}.$$

PROPOSITION 1 (EQUILIBRIUM REPRESENTATION). Under [Assumption 1](#), the market equilibrium exists and is unique.

The base wage vector w and output level y solve the equations

$$(1) \quad w_g = \left(\frac{y}{\ell_g} \right)^{\frac{1}{\lambda}} \Gamma_g(w)^{\frac{1}{\lambda}} \text{ for } g \in \mathbb{G},$$

$$(2) \quad 1 = \left(\Gamma_k(w) + \sum_g \Gamma_g(w) \mu_g(w) w_g^{1-\lambda} \right)^{\frac{1}{1-\lambda}}.$$

In addition, group average wages are given by $\bar{w}_g = w_g \mu_g(w)$.

[Equation \(1\)](#) ensures that base wages clear the labor market. [Equation \(2\)](#) is the ideal price index condition. These equations resemble the equilibrium conditions of an aggregate constant elasticity of substitution (CES) production function that combines capital and labor. The difference is that the CES shares are endogenous and given by task shares, which depend on technology and rents. We denote equilibrium task shares and average group rents by Γ_g and μ_g , respectively.

Our next proposition explains how rents distort equilibrium allocations.

PROPOSITION 2 (INEFFICIENCY). The equilibrium is inefficient:

- (intensive margin) it features too little employment in high-rent tasks;
- (extensive margin) it involves inefficient automation of tasks for which

$$(3) \quad \frac{w_g \mu_{gx}}{\psi_{gx}} > \frac{1}{\psi_{kx} q_x} > \frac{w_g \mu_g}{\psi_{gx}}.$$

Efficiency requires that VMPLs be equalized across tasks within a group (intensive margin), and that this common VMPL exceed what workers could generate in automated tasks (extensive margin). Worker rents distort both margins.⁸

At the intensive margin, firms hire workers until the VMPL equals the task wage w_g . As a result, the VMPL in high-rent tasks exceeds the VMPL in lower-rent tasks. The distortion at the intensive margin depends on the dispersion of rents across tasks within groups, as emphasized in the misallocation literature (e.g., [Hsieh and Klenow 2009](#)).

The extensive-margin inefficiency is more novel: tasks for which [inequality \(3\)](#) holds are inefficiently automated and allocated to capital. Reallocating g workers from nonautomated tasks to producing one unit of task x raises output by $\frac{1}{\psi_{kx} q_x} - \frac{w_g \mu_g}{\psi_{gx}}$ (the value of task x minus workers' opportunity cost). Firms automate these tasks inefficiently because the wage they face is inflated by rents and exceeds workers' opportunity cost. This inefficiency arises whenever the wage for a task that can be automated exceeds what workers could earn elsewhere in the economy and can persist even when rents are identical among all tasks left to workers.

II.C. Automation

We now study how automation affects wages and productivity. We start with the invention of new automation technologies driven by an exogenous increase in q_x from zero to a positive level

8. Apart from worker rents, there are no distortions in this economy. The market equilibrium is therefore efficient if rents are absent (as in [Acemoglu and Restrepo 2022b](#)) or when $\mu_{gx} = \mu_g$ for all $g \in \mathbb{G}$ and $x \in \mathcal{T}$.

$q'_x > 0$ for tasks in $\mathcal{A}_g^T$ in $\mathcal{T}_g$ across $g \in \mathbb{G}$. The sets $\{\mathcal{A}_g^T\}_{g \in \mathbb{G}}$ contain tasks that were initially infeasible to automate but can now be automated thanks to new technologies. For example, advances in robotics in the 1980s and 1990s made it possible to automate industrial tasks such as welding and painting, previously performed by blue-collar workers. The development of enterprise software systems made it feasible to automate clerical tasks performed by clerks and assistants.

1. Endogenous Targeting of High-Rent Tasks. Following the arrival of new opportunities for automating tasks in $\mathcal{A}_g^T$, firms endogenously decide which tasks to automate and which ones to continue producing with labor. We first study this question in partial equilibrium (taking initial base wages as given) and then turn to the equilibrium implications of these choices, which includes how wages change in response to automation.

From cost minimization, firms choose to automate all tasks in

$$(4) \quad \mathcal{A}_g = \left\{ x \in \mathcal{A}_g^T : \frac{w_g \mu_{gx}}{\psi_{gx}} \geq \frac{1}{q'_x \psi_{kx}} \right\}.$$

Intuitively, this set contains the tasks for which the cost of production using the new automation technologies is less than the cost of producing with labor (inclusive of rents).

The set of automated tasks $\mathcal{A}_g$ depends on what technology permits (assumed exogenous and summarized by $\mathcal{A}_g^T$ and q'_x) and the endogenous targeting of high-wage tasks: among the entire set of tasks that can be automated, firms have a stronger incentive to target high-rent tasks. This is because by automating these tasks, firms save more in terms of labor costs, due to the higher wage, $w_g \mu_{gx}$.

To isolate the role of the endogenous targeting of high-rent tasks, which is at the root of all of our remaining results, we first consider a benchmark scenario where opportunities for automation are “neutral.” This is important, since if the opportunities for automation were biased toward high- or low-rent tasks, this bias could drive what is automated in equilibrium. By choosing a neutral configuration, we focus on the forces that guide the endogenous automation of tasks in our model.

ASSUMPTION 2 (NEUTRAL INVENTIONS). Let $F_g(\mu|S)$ denote the “counter CDF” (cumulative distribution function) correspond-

ing to the share of group g employment in tasks in S that pay rents above μ . Then for each group $g \in \mathbb{G}$, we have:

- (i) the rent distribution among tasks that can be newly automated is the same as in all tasks:

$$F_g(\mu \mid \mathcal{A}_g^T) = F_g(\mu \mid \mathcal{T}_g) \quad \forall \mu \geq 1;$$

- (ii) among tasks that can be automated, rents are independent of task-level productivities:

$$F_g(\mu \mid \mathcal{A}_g^T, q'_x \psi_{kx} \leq a, \psi_{gx} \leq b) = F_g(\mu \mid \mathcal{A}_g^T) \\ \forall \mu \geq 1, a > 0, b > 0.$$

The assumption imposes neutrality in the sense that in each group g , opportunities for automation are independent of rents and independent of comparative advantage. These conditions are imposed in groups, so that some groups of workers can still be more exposed to advances in automation than others. We also assume throughout this section that the sets $\mathcal{A}_g^T \setminus \mathcal{A}_g$ and $\mathcal{A}_g$ have positive measure, so that some but not all tasks in $\mathcal{A}_g^T$ are automated, which means there is a meaningful adoption decision. Given the neutrality assumption, any bias in which tasks are automated will be entirely driven by the economic forces in our model, as shown in the next proposition.

PROPOSITION 3 (TARGETING OF HIGH-RENT TASKS). For $g \in \mathbb{G}$, suppose $\mathcal{A}_g^T \setminus \mathcal{A}_g$ and $\mathcal{A}_g$ have positive measure and [Assumption 2](#) holds. Then,

$$F_g(\mu \mid \mathcal{A}_g) > F_g(\mu \mid \mathcal{T}_g) \quad \forall \mu > 1,$$

meaning that the rent distribution in $\mathcal{A}_g$ first-order stochastically dominates the rent distribution in $\mathcal{T}_g$.

The proposition establishes that when opportunities for automation are neutral, endogenous adoption decisions are tilted toward high-rent tasks. Although [Assumption 2](#) provides sufficient conditions for this result, one can see that even with nonneutral inventions, adoption may still be biased toward high-rent tasks. Naturally, this requires that automation possibilities are not too skewed toward low-rent tasks.

[Proposition 3](#) has two important implications. The first is that automation dissipates worker rents. Because adoption endogenously targets high-rent tasks, it reduces the average rents earned by workers. The second is that automation compresses

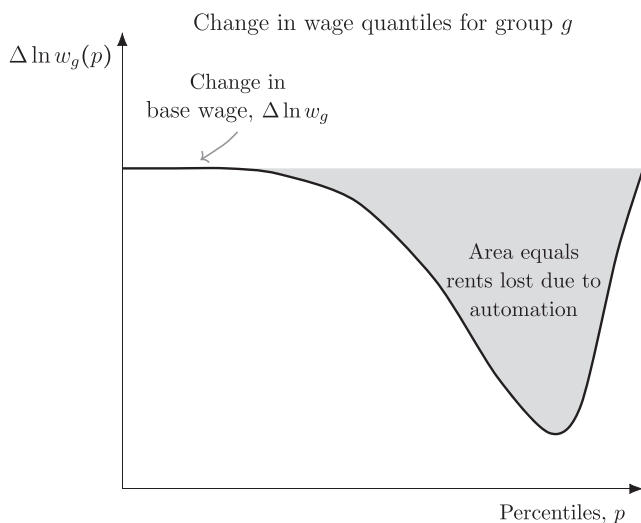


FIGURE I

Predicted Changes in Within-Group Wage Percentiles Due to Automation

wages within exposed groups. By displacing workers out of high-rent tasks, it reduces within-group inequality and generates a distinctive U-shaped pattern of wage changes inside exposed groups.

To state this second implication formally, assume each worker performs a single task, so that the wage distribution across workers in a group is well-defined, and let $\ln w_g(p)$ denote the p th quantile of the log wage distribution in group g (with $p \in [0, 1]$).

PROPOSITION 4 (AUTOMATION AND WITHIN-GROUP WAGE COMPRESSION). For $g \in \mathbb{G}$, suppose $F_g(\mu | \mathcal{A}_g) > F_g(\mu | \mathcal{T}_g)$ for all $\mu > 1$ and let $M_g > 0$ measure group g employment in no-rent tasks. The automation of tasks in $\mathcal{A}_g$ has the following effects:

- For $p \in [0, M_g]$, $\Delta \ln w_g(p) = \Delta \ln w_g$, where $\Delta \ln w_g$ gives the change in base wages.
- For $p \in (M_g, 1)$, $\Delta \ln w_g(p) < \Delta \ln w_g$, which reflects the loss of rents at higher percentiles.
- Suppose moreover that a positive mass of tasks in $\mathcal{T}_g$ cannot be automated and [Assumption 2\(i\)](#) holds. Then, at the top, $\Delta \ln w_g(1) = \Delta \ln w_g$.

[Figure I](#) depicts the U-shaped pattern of within-group wage changes described in the proposition. Workers at the lowest per-

centiles hold tasks with no rents. They earn the base wage w_g , and their wages change only through adjustments in the base wage for group g in equilibrium. Above M_g , wages fall more sharply due to the loss of high-rent jobs, which are more often automated. At the top, two scenarios are possible: either $\Delta \ln w_g(1) < \Delta \ln w_g$ or $\Delta \ln w_g(1) = \Delta \ln w_g$. The second case arises when some of the highest-rent tasks assigned to g cannot be automated.

The results in [Proposition 4](#) offer new empirical predictions, as they contrast with the common view that technological progress increases inequality in a fractal way—between and within groups. Our theory predicts that automation can reduce within-group inequality for exposed groups and identifies this outcome as a tell-tale sign of rent dissipation. This novel implication motivates our first empirical strategy for measuring rent dissipation.

2. Equilibrium Effects. We characterize the equilibrium effects of automation on wages and productivity. We derive formulas for the first-order effects, assuming the sets $\{\mathcal{A}_g^T\}_{g \in \mathbb{G}}$ are small and lie in the interior of $\mathcal{T}_g$.⁹

We describe the effects of automation in three steps. First, we take the set of new automation possibilities $\{\mathcal{A}_g^T\}_{g \in \mathbb{G}}$ and the underlying changes in $\{q'_x\}_{x \in \mathcal{A}_g^T}$ as given. Second, we compute the set of automated tasks $\{\mathcal{A}_g\}_{g \in \mathbb{G}}$ using [equation \(4\)](#) at the initial equilibrium wages. These sets determine the following key objects—all computed at the initial equilibrium allocation—that are sufficient to characterize the effects of automation:

- The task displacement experienced by group g :

$$\delta_g = \frac{\int_{\mathcal{A}_g} \ell_{gx} dx}{\int_{\mathcal{T}_g} \ell_{gx} dx}.$$

This is the reduction in g 's task share due to automation of tasks in $\mathcal{A}_g$ and is given by the share of group g employment in these tasks at the initial allocation.

9. Formally, we assume $\mathcal{A}_g^T$ has measure less than ϵ for some small $\epsilon > 0$ and apply the appropriate notion of “differential” given in [Acemoglu, Kong, and Restrepo \(2024\)](#). With rents, additional technical conditions are required. These and the full derivations appear in the [Online Appendix](#).

- The average cost savings from automating tasks in $\mathcal{A}_g$:

$$\pi_g = \frac{\int_{\mathcal{A}_g} \ell_{gx} \mu_{gx} \pi_{gx} dx}{\int_{\mathcal{A}_g} \ell_{gx} \mu_{gx} dx},$$

where π_{gx} denotes the cost savings from automating task x in $\mathcal{A}_g$:

$$\pi_{gx} = \frac{1}{\lambda - 1} \left[\left(\frac{q'_x \psi_{kx} w_g \mu_{gx}}{\psi_{gx}} \right)^{\lambda-1} - 1 \right] \geq 0.$$

- The average rent earned by group g workers in newly automated tasks:

$$\mu_{\mathcal{A}_g} = \frac{\int_{\mathcal{A}_g} \ell_{gx} \mu_{gx} dx}{\int_{\mathcal{A}_g} \ell_{gx} dx}.$$

Although we provide sufficient conditions under which automation targets high-rent tasks and identify economic forces ensuring $\mu_{\mathcal{A}_g} > \mu_g$, the characterization of automation's effects below does not rely on [Assumption 2](#). We invoke this assumption only when a sufficient condition is needed to guarantee $\mu_{\mathcal{A}_g} > \mu_g$.

The quantities $\{\{\delta_g\}_{g \in \mathbb{G}}, \{\pi_g\}_{g \in \mathbb{G}}, \{\{\mu_{\mathcal{A}_g}\}_{g \in \mathbb{G}}\}\}$ summarize the direct impact of new automation technologies on task shares, costs, and average group rents holding the allocation of nonautomated tasks constant.

The final step incorporates the reassignment of marginal tasks across groups in response to these direct effects, tracing the resulting ripple effects. [Figure II](#) illustrates both direct and ripple effects: the left panel shows the initial allocation of tasks to g, g' , and capital, and the right panel introduces the new automation technologies $\mathcal{A}_g^T$ and the subset of tasks automated, $\mathcal{A}_g$. The displacement of g workers from $\mathcal{A}_g$ triggers an endogenous reassignment of tasks from other factors toward g , as its relative base wage declines. These ripple effects are depicted with the dashed curves, indicating how tasks are reallocated.

To characterize the consequences of ripple effects, we follow [Acemoglu and Restrepo \(2022b\)](#) and differentiate [equation \(1\)](#) to obtain:

$$d \ln w_g = \frac{1}{\lambda} d \ln y - \frac{1}{\lambda} \delta_g + \frac{1}{\lambda} \sum_{g'} \frac{\partial \ln \Gamma_g(w)}{\partial \ln w_{g'}} d \ln w_{g'} \quad \text{for } g \in \mathbb{G}.$$

The term δ_g represents the task displacement from automation experienced by group g , and the third term captures ripple effects.

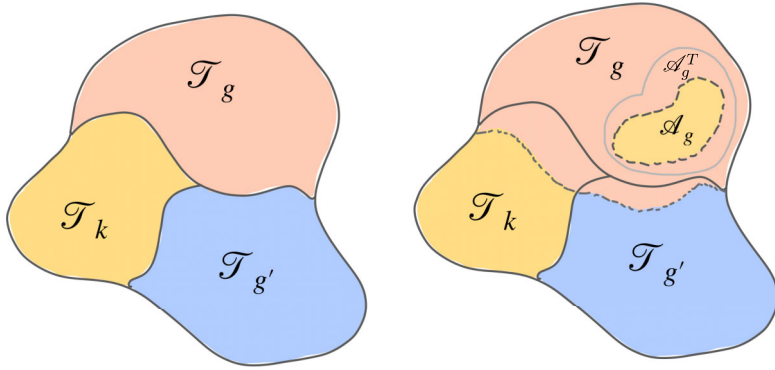


FIGURE II

Task Allocation and the Effects of Automation

The left panel shows the assignment of tasks across factors in the initial equilibrium. The right panel illustrates new automation possibilities, the subset of tasks automated, and the resulting task displacement and ripple effects.

Stacking these equations and solving for $d \ln w_g$, we obtain:

$$d \ln w_g = \frac{1}{\lambda} \Theta_g \text{stack} (d \ln y - \delta_j), \quad \text{with } \Theta = \left(\mathbb{1} - \frac{1}{\lambda} \mathcal{J}_\Gamma \right)^{-1}.$$

Here $\text{stack}(x_j)$ denotes the column vector $(x_1, x_2, \dots, x_G)$, and $\mathcal{J}_\Gamma$ is the $G \times G$ Jacobian with entries $\mathcal{J}_{\Gamma, g, g'} = \frac{\partial \ln \Gamma_g(w)}{\partial \ln w_{g'}}$. As in [Acemoglu and Restrepo \(2022b\)](#), we refer to Θ as the propagation matrix (because it propagates an initial direct impact across the economy). Each entry $\theta_{gg'}$ is nonnegative and captures the extent to which shocks that reduce the base wage of group g' affect group g via ripple effects.¹⁰

In the presence of distortions, we also need to keep track of how ripple effects influence average group rents. This information is summarized by the rent-impact matrix:

$$\mathcal{M} = \mathcal{J}_\mu \left(\mathbb{1} - \frac{1}{\lambda} \mathcal{J}_\Gamma \right)^{-1},$$

10. The propagation matrix takes the form of a Leontief inverse because it accumulates successive rounds of task reassignment. This inverse exists and has positive entries, and its eigenvalues are in $[0, 1]$. This means that ripple effects play an equalizing role and dampen the direct effects of automation. The entries of the propagation matrix also summarize how substitutable groups of workers are in the aggregate.

where $\mathcal{J}_\mu$ is a $G \times G$ Jacobian with entries $\mathcal{J}_{\mu, g, g'} = \frac{\partial \ln \mu_g(w)}{\partial \ln w_{g'}}$. The entries in this matrix indicate whether competition between group g' and g occurs at tasks where g workers earn above-average rents (positive entries) or below-average rents (negative entries).

The following proposition provides our formulas for the first-order equilibrium effects of new automation technologies on wages, rents, and productivity.

PROPOSITION 5 (EQUILIBRIUM EFFECTS OF AUTOMATION). Consider new automation technologies represented by (small interior) sets $\{\mathcal{A}_g^T\}_{g \in \mathbb{G}}$. Let $\{\mathcal{A}_g\}_{g \in \mathbb{G}}$ be given by [equation \(4\)](#) and the corresponding direct effects of automation be given by $\{\delta_g\}_{g \in \mathbb{G}}, \{\pi_g\}_{g \in \mathbb{G}}, \{\{\mu_{\mathcal{A}_g}\}_{g \in \mathbb{G}}\}$. The first-order equilibrium effects of automation on base wages and output are given by the formulas

$$(5) \quad d \ln w_g = \Theta_g \text{stack} \left(\frac{1}{\lambda} d \ln y - \frac{1}{\lambda} \delta_j \right) \quad \text{for } g \in \mathbb{G}$$

$$(6) \quad \sum_g s_g d \ln w_g = \sum_g s_g \delta_g \frac{\mu_{\mathcal{A}_g}}{\mu_g} \pi_g,$$

where s_g is the share of g 's earnings in output. Moreover, the change in group rents is

$$(7) \quad d \ln \mu_g = \mathcal{M}_g \text{stack} \left(\frac{1}{\lambda} d \ln y - \frac{1}{\lambda} \delta_j \right) - \left(\frac{\mu_{\mathcal{A}_g}}{\mu_g} - 1 \right) \delta_g.$$

and the change in aggregate consumption is $d \ln c = \frac{1}{s_\ell} d \ln \text{TFP}$, where s_ℓ is the labor share in output and $d \ln \text{TFP}$ is the change in TFP from automation:

$$(8) \quad d \ln \text{TFP} = \sum_g s_g \delta_g \frac{\mu_{\mathcal{A}_g}}{\mu_g} \pi_g + \sum_g s_g d \ln \mu_g.$$

Proof. We sketch the proof and provide details in the [Online Appendix](#). [Equation \(5\)](#) was derived above. [Equation \(6\)](#) follows from differentiating the price index condition in [equation \(2\)](#). Together with [equation \(5\)](#), it forms a system of equations that pins the change in output.

Define $d \ln \text{TFP} = d \ln y - s_K d \ln k$. With constant returns to scale, we obtain the dual version of the Solow residual: $d \ln \text{TFP} = \sum_g s_g (d \ln w_g + d \ln \mu_g)$. Substituting $\sum_g s_g d \ln w_g$ from [equation \(6\)](#) yields [equation \(8\)](#). The fact that $d \ln c = \frac{1}{s_\ell} d \ln \text{TFP}$ follows from the resource constraint $c = y - k$. $\square$

Using the formulas in the proposition, we obtain the impact of automation on average group wages $\bar{w}_g = w_g \mu_g$ as follows:

$$(9) \quad d \ln \bar{w}_g = (\Theta_g + \mathcal{M}_g) \text{stack} \left(\frac{1}{\lambda} d \ln y - \frac{1}{\lambda} \delta_j \right) - \left(\frac{\mu_{\mathcal{A}_g}}{\mu_g} - 1 \right) \delta_g.$$

To gain intuition for this equation, consider an economy where groups can only produce disjoint tasks and capital produces all tasks with $q_x > 0$, so there are no ripple effects, the propagation matrix is the identity, and all of the entries of the rent-impact matrix are zero.¹¹ Equation (9) then becomes:

$$(10) \quad d \ln \bar{w}_g = \underbrace{\frac{1}{\lambda} d \ln y}_{\text{productivity effect}} - \underbrace{\frac{1}{\lambda} \delta_g}_{\text{displacement effect}} - \underbrace{\left(\frac{\mu_{\mathcal{A}_g}}{\mu_g} - 1 \right) \delta_g}_{\text{rent dissipation}}.$$

Without ripple effects, the change in average group wages depends on three forces. The first two operate in models with competitive labor markets, such as Acemoglu and Restrepo (2022b), and capture the effects of automation working via labor demand and base wages. These include a positive productivity effect (cost reductions due to automation expand output and increase the demand for all workers) and a negative displacement effect (automation reduces the share of tasks employing workers from group g by δ_g).

The third term captures the new rent dissipation channel. This force amplifies wage losses for exposed groups whenever $\mu_{\mathcal{A}_g} > \mu_g$ —because automation displaces workers from higher-rent tasks and pushes them into lower-wage jobs. As noted, Assumption 2 is sufficient but not necessary for this pattern. In general, the strength of rent dissipation depends on the magnitude of $\frac{\mu_{\mathcal{A}_g}}{\mu_g}$ —how high average rents were in newly automated tasks relative to the average.

When ripple effects are present in equation (9), group g 's average wages additionally depend on whether other groups competing with it are displaced by automation. The propagation matrix contains all the information about how these ripple effects work, and the rent-impact matrix adjusts for how this reassignment changes group-level average rents.

One important difference between the displacement effect and rent dissipation is in their propagation. Automation reduces

11. Formally, one can shut down ripple effects by assuming that (i) for all x , $\psi_{gx} > 0$ implies $\psi_{g'x} = 0$ for all $g' \neq g$; and (ii) for all x , $\psi_{kx} > 0$ implies $\psi_{kx} > \underline{\psi}_k$ for some threshold $\underline{\psi}_k > 0$.

the (relative) base wage of exposed groups, and this enables exposed groups of workers to gain some marginal tasks, propagating the incidence of the shock to others and dampening its initial adverse effects on them. This is the reason the effects of task displacement are mediated by the matrices Θ and $\mathcal{M}$. In contrast, rent dissipation works by reducing group wages in excess of base wages, and does not generate additional task reassignments. Consequently, exposed groups bear the full incidence of rent dissipation, which does not propagate via Θ and $\mathcal{M}$.

Next we discuss the implications of automation for productivity and welfare. In our economy, changes in TFP are proportional to changes in consumption and the average change in wages paid to workers: $d \ln \text{TFP} = s_\ell d \ln c = \sum_g s_g d \ln \bar{w}_g$. It is therefore sufficient to derive the impact of automation on TFP, from which the effects on consumption and average wage levels can be seen.

In the absence of ripple effects, the formula for changes in TFP in [equation \(8\)](#) simplifies to

$$(11) \quad d \ln \text{TFP} = \underbrace{\sum_g s_g \delta_g \frac{\mu_{\mathcal{A}_g}}{\mu_g} \pi_g}_{\text{direct gains à la Hulten}} - \underbrace{\sum_g s_g \left(\frac{\mu_{\mathcal{A}_g}}{\mu_g} - 1 \right) \delta_g}_{\text{changes in allocative efficiency}}.$$

This formula relates to [Baqae and Farhi \(2020\)](#), who decompose the impact of technology in inefficient economies into a direct effect plus changes in allocative efficiency. The first term in [equation \(11\)](#) represents the direct benefits from reducing the cost of producing automated tasks. This term is positive and has the same envelope logic as Hulten's theorem in competitive economies: it is the product of (i) the cost share of automated tasks in output, given by their Domar weights, $s_g \delta_g \frac{\mu_{\mathcal{A}_g}}{\mu_g}$; and (ii) average cost savings from automation, $\pi_g > 0$. The second term, in turn, reflects changes in allocative efficiency from automation. This term captures how changes in the allocation of labor across tasks affect output. In a competitive economy, an envelope argument implies that this term is zero—the initial allocation of labor already maximized output and thus small changes only have second-order consequences. This is no longer true in the presence of labor market distortions.

A key observation in our economy is that automating jobs that pay above-average rents ($\mu_{\mathcal{A}_g} > \mu_g$) worsens allocative efficiency. At first this may seem paradoxical, since one might expect that eliminating high-rent jobs improves efficiency. This intuition

is incorrect. Automation reallocates labor from high-rent to low-rent jobs, where workers' VMPL is lower. By pushing workers into lower-VMPL jobs, automation reduces allocative efficiency.¹²

A complementary interpretation of the TFP formula in equation (11) views the first term as firms' and consumers' willingness to pay for the new technology. This value is positive and equals the consumer surplus from reallocating tasks in $\mathcal{A}_g$ from labor (produced at cost $\frac{w_g \mu_{gx}}{\psi_{gx}}$) to capital (produced at the lower cost $\frac{1}{q'_x \psi_{kx}}$). This surplus, however, overstates the social value of new automation technologies because it ignores the first-order cost to workers of losing high-rent jobs, captured by the rent dissipation term.¹³

The TFP formula in equation (11) establishes that rent dissipation offsets part of the productivity gains from automation and can even reduce aggregate productivity, average real wages, and welfare. This happens when new automation technologies are adopted in tasks where workers earn high rents but yield only small cost savings, so that automation saves on rents more than it improves efficiency. By contrast, in competitive labor markets automation always raises TFP, consumption, and average wages (even if it reduces the real wages of exposed groups), because allocative efficiency is unaffected.

In the general case with ripple effects, the TFP expression in equation (8) shows that allocative efficiency may deteriorate either through rent dissipation (as discussed above) or because ripple effects reallocate workers away from high-rent tasks. This second force is summarized by the matrix $\mathcal{M}$ in the term $\sum_g s_g d \ln \mu_g$.

This discussion also clarifies the role of fixed wedges in our theory. Rent dissipation is inefficient because automation displaces workers from high-rent tasks, which occurs due to the pres-

12. Not all forms of technological progress reduce allocative efficiency. An improvement in capital productivity in tasks already assigned to this factor (e.g., a better automatic drill or photocopier) has no direct effect on allocative efficiency. Nor do labor-augmenting technological advances at the intensive margin (meaning improvements in productivity for a group in tasks already assigned to that group). The fact that automation operates at the extensive margin, shifting high-rent tasks away from labor, is responsible for its direct impact on allocative efficiency.

13. In our model, firms' and consumers' willingness to pay coincide because there are no product market distortions. If labor market rents were linked to producer markups, automation would also affect allocative efficiency through its impact on the economy's product mix.

ence of fixed wedges. More generally, the same inefficiency would arise even if wedges adjusted partially, so long as they did not do so fully and thus ensure an efficient level of hiring and separations. Our empirical analysis in the next section shows that automation has indeed shifted U.S. workers away from high-rent jobs, supporting the view that wedges do not fully adjust and that automation inefficiently dissipates rents.

II.D. Incorporating Additional Within-Group Differences

Our base model assumes that all workers in a demographic group are identical in terms of their absolute and comparative advantages, and the only source of wage differences within groups is due to rents. In practice, there are additional sources of within-group heterogeneity, and we now show how two of these—within-group productivity differences and compensating differentials—can be incorporated into our framework. These extensions show that such forms of heterogeneity do not alter our key results, and they also provide guidance on how empirical work can control for these differences. The details for both extensions are in [Online Appendix S4](#).

1. Within-Group Productivity Differences. In our baseline model, all workers in group g have identical task-level productivities. We now relax this by assuming that workers from group g have productivity $\psi_{gx} z$ in task x , where z is an individual-specific productivity level. This formulation assumes that workers in a group share the same comparative advantage, but differ in their absolute advantages, given by their productivity level z .

This generalization affects only within-group wages. A worker from group g with individual productivity z performing task x is paid

$$w_{gx}(z) = w_g \mu_{gx} z,$$

where w_g is the base wage for workers with $z = 1$. This wage equation follows from firms' indifference across workers of different skill levels, which equates wages per efficiency unit within each group. Put differently, this wage rule ensures that the cost of producing one unit of task x with group g labor is $\frac{w_g \mu_{gx}}{\psi_{gx}}$, exactly as in the baseline model, regardless of the individual productivity z of the worker performing that task.

Given this wage rule, the equilibrium mirrors the baseline model, except that there are now within-group wage differences

as well. Firms take wages as given and allocate tasks to minimize costs, and this leads to the same assignment of tasks to factors as before.

The only complication arises from rationing, which is again present since only some workers will get the high-rent jobs. Rationing can now condition on individual skill (though there is no reason for firms to prefer to do so, since they make the same profits from all workers in the same demographic group). We assume that the rationing protocol preserves ranks, so that rents do not alter relative wages in a group. That is, for any two workers from group g with individual skills $z > z'$, we have $w_{gx}(z) > w_{gx}(z')$.

The [Online Appendix](#) shows that all our theoretical results remain valid in this extension. In particular, automation still targets high-rent tasks regardless of the skill level of the workers performing them, since adoption decisions depend only on wages per efficiency unit of labor, which are the same as in the baseline model. This targeting generates the same U-shaped pattern for wage changes described in [Proposition 4](#) and [Figure I](#), because differences in z cancel out with our rationing rule.

The key implication of this extension is that there may be level differences in productivity and wages across workers within the same group, including possibly across the lower percentiles of the within-group wage distribution. Nevertheless, all our results and the U-shaped pattern depicted in [Figure I](#) for wage changes are unaffected.

2. Compensating Differentials. The baseline model ignores the possibility that wage differences may reflect compensating differentials. Some jobs may pay more than others because they are unpleasant or difficult to perform. To capture this, we assume that workers from group g derive utility

$$u_{gx} = \frac{w_{gx}}{\zeta_{gx}}$$

from employment in task x . The numerator represents utility from consumption, which depends on the wage w_{gx} . The denominator is a disutility from performing task x , denoted by $\zeta_{gx} > 0$. This parameter captures task attributes such as hazardous conditions, long shifts, or monotony, and is allowed to vary across groups and tasks (for example, construction jobs may be hard for all workers, but particularly for older ones). This specification implies that worker utility depends on net wages, given by $\frac{w_{gx}}{\zeta_{gx}}$.

We assume the following wage rule in this extension:

$$w_{gx} = w_g \mu_{gx} \zeta_{gx},$$

where w_g is the base utility level of g workers in jobs that pay no rents, and w_{gx} again denotes the wage of workers from group g in task x . In the [Online Appendix](#), we show that this wage rule follows from our micro-foundations, where net wages must exceed the base utility w_g by a rent μ_{gx} due to either efficiency wages or bargaining.¹⁴

Two important results follow from this extension (both established in the [Online Appendix](#)). First, because wages depend on the composite term $\mu_{gx}\zeta_{gx}$, compensating differentials and rents have the same positive implications, with automation targeting high-rent and/or high-compensating differential jobs. In essence, a job with high ζ_{gx} is more expensive for the employer and thus more profitable to automate. However, these two features have different normative implications. Although automating high-rent jobs can be inefficient, this is not the case for jobs with high compensating differentials. A planner would also choose to automate unpleasant or dangerous jobs, since they impose a greater burden on workers.

Second, given these distinct normative consequences, our empirical analysis pays close attention to separating rents from compensating differentials. In this regard, the extension shows that [Propositions 4](#) and [5](#) now hold for net wages (as defined above), rather than raw wages. This means that when compensating differentials are present, estimates of automation's effect on rents should control for these differentials (or be based on wages purged of their influence). This is the strategy we adopt in the next section.

III. REDUCED-FORM EVIDENCE

This section presents reduced-form evidence on the impact of automation on group-level wages, within-group wage dispersion, and worker rents in the United States between 1980 and 2016.

14. Two additional points: first, measured wages in a group can also vary due to in-kind payments, such as a company car or accommodation. To the extent that these are transfers between firms and workers, they will not affect allocations or incentives for automation. Second, in some models of labor market imperfections, part of the amenity value of a job could be captured by the employer. This possibility would complicate the analysis but would not affect our conclusions.

We focus on 500 detailed demographic groups, defined by five education levels, gender, five age groups, five race and ethnicity groups, and native/immigrant status. We estimate group-level specifications of the form

$$\begin{aligned} &\text{Change in group } g \text{ outcome 1980–2016} \\ (12) \quad &= \beta \text{ Task displacement}_g + X_g \gamma + u_g, \end{aligned}$$

where $\text{Task displacement}_g$ is the empirical analogue of δ_g and measures the task displacement due to automation experienced by group g between 1980 and 2016. In addition, X_g includes covariates, and u_g denotes the residual. When estimating this equation, we weigh groups by their share of U.S. employment in 1980 and report standard errors that are robust to heteroskedasticity.

Our outcomes include changes in group average real hourly wages and changes in the p th percentile of the within-group wage distribution, $\Delta \ln w_g(p)$ for $p = 5, 10, \dots, 95, 99$. These are computed from the 1980 census and by pooling 2015–2017 American Community Survey (ACS) data. We use the cleaning procedures from [Acemoglu and Restrepo \(2022b\)](#).

We also use the Basic Monthly and Displaced Worker Supplement from the Current Population Survey (CPS) to create proxies for worker rents, and the 1977 Quality of Employment Survey to construct proxies for workplace amenities and compensating differentials. This survey asked a cross section of respondents across all U.S. industries about working conditions and job satisfaction. We aggregate answers at the industry level (using the 49 industries in our sample) and create four measures that we use to purge wage differences from compensating wage differentials:

- (i) A measure of *job amenities*, summarizing the extent to which respondents perceive their jobs as having desirable conditions (ease of travel, good hours, generous benefits, autonomy, security, and comfort).
- (ii) A measure of *job meaning*, summarizing the extent to which respondents believe their work is meaningful and important.
- (iii) A measure of *willingness to pay for improved working conditions*, giving the extent to which respondents would take a wage cut to improve conditions in their job.
- (iv) A composite *job quality index*, defined as the first principal component of the three measures above.

III.A. Measuring Task Displacement

Our strategy for measuring task displacement follows [Acemoglu and Restrepo \(2022b\)](#) and is based on the following assumption:

ASSUMPTION 3 (MEASUREMENT ASSUMPTION). During 1980–2016, only routine tasks were automated, and within an industry, automation displaced all groups of workers from routine tasks at a common rate.

[Online Appendix S5.7](#) shows that in the multi-sector version of our model (described in [Section IV](#)) and under [Assumption 3](#), task displacement across groups can be estimated as

$$(13) \quad \text{Task displacement}_g = \sum_i \frac{\ell_{gi}}{\ell_g} \text{RCA}_{gi}^{\text{routine}} \times \frac{-\Delta \ln s_{\ell i}^A}{(1 + s_{\ell i}(\lambda - 1)\pi_i) \frac{\mu_{\mathcal{A}_i}}{\mu_i}}.$$

This measure is the product of three terms:

- (i) The term $\frac{\ell_{gi}}{\ell_g}$ represents group g 's exposure to industry i . It accounts for group g 's specialization in sectors that automated more tasks.
- (ii) The term $\text{RCA}_{gi}^{\text{routine}}$ is a measure of the revealed comparative advantage of group g in routine jobs in industry i . It apportions the incidence of automation in the industry based on who performs routine tasks. This term is the reason, for example, managers and blue-collar workers in the manufacturing industry are not equally affected by the introduction of industrial robots.
- (iii) The term $-\Delta \ln s_{\ell i}^A$ is the percent reduction of the labor share of industry i caused by automation between 1980 and 2016. In our framework, there is a direct relationship between the rate at which tasks are automated in an industry and the resulting decline in its labor share, $-\Delta \ln s_{\ell i}^A$. This term is divided by $1 + s_{\ell i}(\lambda - 1)\pi_i$ to adjust for the effects of automation on the labor share working via substitution across tasks, and by the term $\frac{\mu_{\mathcal{A}_i}}{\mu_i}$ —the average (relative) rent in tasks automated in industry i —to adjust for the effects of automation operating

through worker rents (this last adjustment is not present in competitive markets).

We measure task displacement using data for 49 industries from the BEA Integrated Industry-Level Production Accounts. We compute employment shares by industry, $\frac{\ell_{gi}}{\ell_g}$, and the revealed comparative advantages $RCA_{gi}^{\text{routine}}$ for the 500 demographic groups from the 1980 census. We define routine jobs as the 33% occupations with the highest routine content according to O*NET. As in [Acemoglu and Restrepo \(2022b\)](#), we estimate $-\Delta \ln s_{gi}^A$ as the predicted labor share decline from a cross-industry regression of percent labor share changes by industry (from the BEA, between 1987 and 2016, and rescaled to a 36-year change) against three proxies for automation: the adjusted penetration of industrial robots (from [Acemoglu and Restrepo 2020](#)), the increase in the share of specialized software services in value added, and the increase in the share of dedicated machinery in value added (from the BLS Total Multifactor Productivity Tables). For the adjustment term, we set $\lambda = 0.5$, $\pi_i = 30\%$, and $\frac{\mu_{\mathcal{A}_i}}{\mu_i} = 1.35$ for all industries. These choices are motivated in [Section IV](#).

The left panel of [Figure III](#) summarizes the industry labor share trends. The shaded bars show the observed percent labor share declines. The hollow bars depict the component attributed to our proxies of automation, which jointly explain 50% of cross-industry changes in labor shares since 1987.

The right panel of [Figure III](#) depicts our measure of task displacement from automation over 1980–2016, for 500 U.S. demographic groups, plotted against their baseline hourly wages in 1980 on the horizontal axis. Groups with postcollege degrees saw few of their tasks automated between 1980 and 2016, while workers in the middle and lower middle of the wage distribution lost 15%–20% of their tasks to automation.¹⁵

III.B. The Effect of Automation on Average Group Wages

We examine the reduced-form relationship between automation and average group wages. We estimate [equation \(12\)](#) using

15. The measure in [equation \(13\)](#) is the same as in [Acemoglu and Restrepo \(2022b\)](#), except for the term $\frac{\mu_{\mathcal{A}_i}}{\mu_i}$, since our previous work did not consider the role of rents. There are also minor differences in weights outlined in [Online Appendix S5](#). These explain the differences in magnitudes and estimates between the papers.

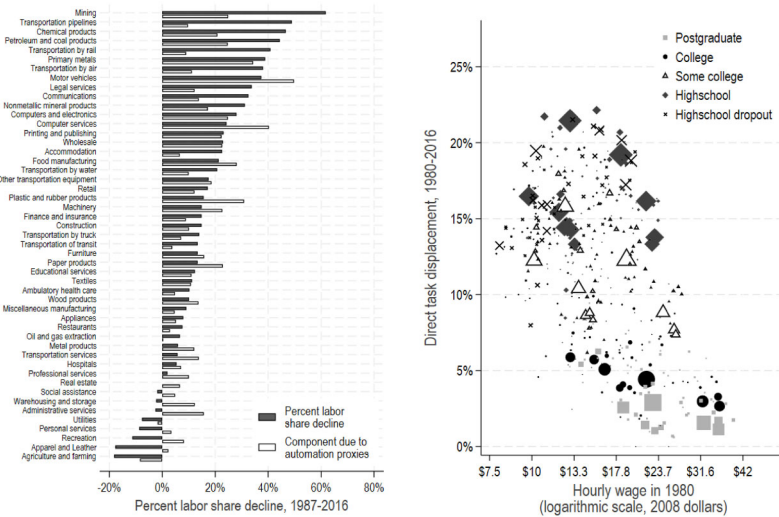


FIGURE III

Task Displacement from Automation Across Industries and Groups

The left panel shows labor share declines across U.S. industries from 1987 to 2016 (positive values indicate declines). Orange bars mark the decline explained by automation, using our three proxies. The right panel reports direct task displacement for 500 demographic groups between 1980 and 2016, as measured by equation (13).

the change in group log average wages between 1980 and 2016 as the dependent variable, following Acemoglu and Restrepo (2022b).

The left panel of Figure IV shows the relationship between changes in average group wages and exposure to automation. The point estimate indicates that a 10 percentage point increase in task displacement is associated with a 24% decline in group-level relative wages. This single measure of exposure explains 66% of the variation in wage changes among U.S. demographic groups since 1980.

The right panel of Figure IV depicts this relationship after controlling for our baseline covariates, including (i) gender and education dummies, which capture other forms of skill-biased technological change affecting college and postcollege groups, as well as social or technological changes affecting women relative to men; and (ii) three measures of changes in the economy's sectoral composition, which influence labor demand and worker rents. These measures are the employment shares of groups in manufacturing

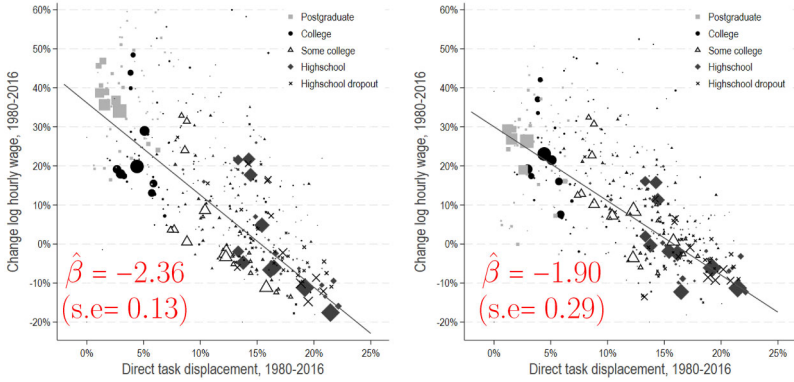


FIGURE IV

Reduced-Form Relationship Between Wage Changes and Task Displacement

The left panel shows the relationship between changes in group wages (in logs) and task displacement. The right panel controls for gender, education, and sectoral demand and rent shifters.

in 1980, which control for shocks affecting this sector;

$$\text{Sectoral demand shifts}_g = \sum_i \frac{\ell_{gi}}{\ell_g} \Delta \ln \text{value added}_i,$$

which accounts for exposure to expanding sectors (in terms of value added); and

$$\text{Sectoral rent shifts}_g = \sum_i \frac{\ell_{gi}}{\ell_g} \left(\frac{\bar{w}_{gi}}{\bar{w}_g} - 1 \right) \Delta \ln \text{value added}_i,$$

which incorporates the effects of expansions (or contractions) of sectors where workers earn high rents. In this expression, $\frac{\bar{w}_{gi}}{\bar{w}_g}$ is the ratio between the average wage earned by a group in industry i and that group's average wage.¹⁶

The point estimate in the right panel is $\hat{\beta} = -1.9$, implying that a 10 percentage point increase in task displacement from automation is associated with a 20% decline in average group wages. With these additional controls, differences in exposure to automation explain 53% of the variation in wage changes among U.S. demographic groups since 1980.

16. For these covariates, we measure $\frac{\ell_{gi}}{\ell_g}$ and $\frac{\bar{w}_{gi}}{\bar{w}_g}$ from the 1980 census, and the change in value added from the BEA industry accounts for 1987–2016.

III.C. Rent Dissipation

The relationships shown in [Figure IV](#) are similar to the results reported in [Acemoglu and Restrepo \(2022b\)](#). Our framework, however, recognizes that this relationship reflects both the direct displacement effects of automation (present even in competitive labor markets) and rent dissipation (which arises when labor market rents exist and automation targets high-rent tasks). The goal of this section is to decompose this relationship into these two components, using two complementary strategies.

The first strategy exploits within-group wage changes associated with automation. As implied by our theory, these changes follow a U-shaped pattern, which we use to infer the magnitude of rents in jobs displaced by automation. The second strategy relies on explicit proxies for rents at the industry and occupation levels to show that automation shifts exposed groups of workers away from high-rent jobs. In both strategies, we pay special attention to controlling for the role of compensating differentials.

1. Estimates of Rent Dissipation from Within-Group Wage Changes. In our theory, automation affects wages at the bottom percentiles of the within-group distribution exactly by the change in the base wage, $\Delta \ln w_g$, since these workers earn no rents. Any further wage declines beyond this level reflect the loss of rents, as workers are displaced from high-rent tasks. We can estimate the contribution of rent dissipation by comparing the larger wage declines at higher within-group percentiles to those at lower percentiles.

We implement this strategy by estimating a variant of [equation \(12\)](#) with the dependent variable as the change in log wages at the p th percentile of the within-group wage distribution between 1980 and 2016, $\Delta \ln w_g(p)$, for $p = 5, 10, \dots, 95, 99$:

$$(14) \quad \Delta \ln w_g(p) = \beta(p) \text{Task displacement}_g + X_g \gamma(p) + u_g^p.$$

This is an unconditional group-quantile regression, as in [Chetverikov, Larsen, and Palmer \(2016\)](#), which estimates the effect of automation at each percentile.

[Figure V](#) plots our estimates of $\beta(p)$ from a parsimonious specification that controls only for sectoral shifts. A clear U-shape is visible. Groups exposed to automation experienced a more pronounced decline in wages between the 70th and 90th percentiles of the within-group distribution, with estimates of $\beta(p)$ ranging from -2.5 to -3 . This implies that wages at these percentiles fell

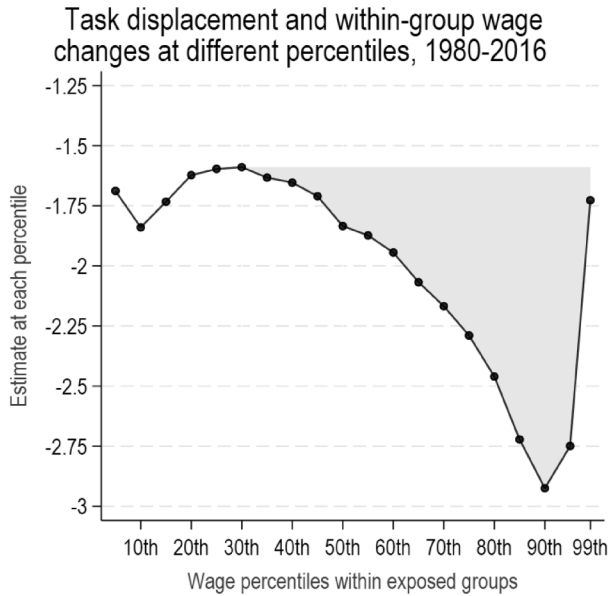


FIGURE V

Reduced-Form Relationship Between Within-Group Wage Changes and Task Displacement

The figure shows estimates from an unconditional quantile regression of $\Delta \ln w_g(p)$ on task displacement for percentiles p from the 5th to the 99th (see [equation \(14\)](#)), controlling for sectoral shifts.

by 25%–30% for every 10 percentage point increase in task displacement. In contrast, the estimates of $\beta(p)$ are fairly uniform between the 5th and 40th percentiles, ranging from -1.6 to -1.75 . Wages at these bottom percentiles fell by only 16% for every 10 percentage point increase in task displacement.

This is the distinctive pattern predicted by our theory. In light of [Proposition 4](#), workers below the 30th (or the 40th) percentiles of the within-group wage distribution earn no rents, and the common decline in wages below the 30th or 40th percentile reflects the fall in group base wages via equilibrium effects. The more pronounced declines between the 50th and 95th percentiles (shown by the gray-shaded area) capture the loss of rents. We also see that wages at the 99th percentile fall by a similar amount as those at the bottom. This is what the proposition predicts in the natural

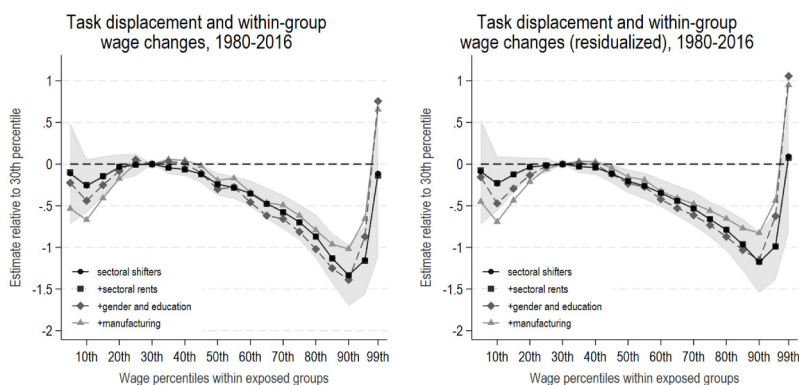


FIGURE VI

Robustness of the Reduced-Form Relationship between Within-Group Wage Changes and Task Displacement

The figure shows estimates from an unconditional quantile regression of $\Delta \ln w_g(p)$ on task displacement for percentiles p from the 5th to the 99th (equation (14)), relative to the 30th percentile. The left panel uses observed wages, while the right panel uses wages purged of job-quality measures.

case where some high-rent tasks performed by exposed groups remain impossible to automate.¹⁷

Figure VI demonstrates the robustness of the U-shaped pattern. The left panel reports estimates of $\beta(p)$ relative to the 30th percentile, showing how much wages decline beyond this base level. The black line shows estimates from our baseline specification, which controls only for sectoral shifts. The remaining lines add controls for sectoral rent shifts, gender and education dummies, and exposure to manufacturing. The last specification is conservative, since part of what we seek to explain is the loss of high-rent jobs in manufacturing. In all specifications, wages decline by similar amounts between the 5th and 40th percentiles (the differences below the 30th percentile, when controlling for manufacturing, are not statistically different from zero) and show more pronounced declines at higher percentiles (except at the very top).

The interpretation of the U-shape as reflecting rent dissipation due to automation does not require us to assume away other sources of within-group wage differences. As discussed in

17. This refers to the 99th percentile of the within-group wage distribution, not the overall wage distribution. The upward-sloping portion of the U-shape is therefore not driven by top-coded census wages.

Section II.D, even if workers differ in productivity levels, automation would still generate the same U-shaped pattern of wage changes. Although workers at different percentiles differ in productivity and thus baseline wage levels, the quantile regression in [equation \(14\)](#) continues to identify changes in base wages and rents, provided automation does not alter the ranking of workers within a group.

The U-shape pattern is also not explained by compensating wage differentials. The right panel of [Figure VI](#) shows that the same U-shape is visible for net wage changes. To compute net wages, we use the job-quality index from the 1977 Job Quality Survey. In a first stage, we regress wages on job quality to purge wages of compensating differentials. We reestimate [equation \(14\)](#) using net (or residual) wage changes at each percentile of the within-group distribution as the dependent variable. The resulting pattern is similar, though slightly less pronounced, suggesting that the bulk of the U-shape is driven by the loss of rents rather than compensating differentials.

Motivated by the robust U-shaped pattern, our first strategy for measuring rent dissipation decomposes the average wage change of exposed groups, $\Delta \ln \bar{w}_g$, into two components: a base wage change, given by the decline at the 30th percentile, $\Delta \ln w_g(30\text{th})$, and a rent-dissipation component, given by the more pronounced decline above the 30th percentile, $\Delta \ln \bar{w}_g - \Delta \ln w_g(30\text{th})$, attributed to automation. This latter component corresponds to the shaded area in [Figure V](#).

[Figure VII](#) presents estimates of [equation \(12\)](#), using the two components $\Delta \ln w_g(30\text{th})$ and $\Delta \ln \bar{w}_g - \Delta \ln w_g(30\text{th})$ as dependent variables. Both specifications control for baseline covariates. The left-most panel shows estimates of the decline in base wages. A 10 percentage point increase in task displacement is associated with a 15.3% decline in the base wages of exposed groups. The remainder of the average wage effect is due to rent dissipation, shown in the middle panel. A group with 10 percentage points more task displacement experiences an additional 3.7% decline in wages above the 30th percentile, which our theory interprets as the loss of rents from automation. The point estimate of $\beta = -0.37$ (std. err. = 0.11) implies a rent-dissipation rate of 37% (corresponding to $\frac{\mu_{Ag}}{\mu_g} - 1$ in the theory). Overall, this means that one-fifth of the average wage decline in [Figure IV](#) is due to rent dissipation, with the rest driven by base-wage changes.

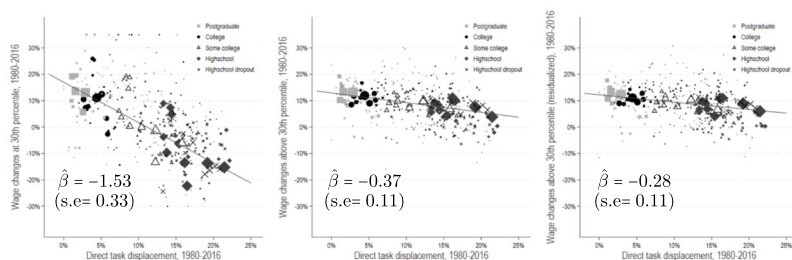


FIGURE VII

Reduced-Form: Changes in Base Wages and Rents from Automation

The left panel plots the relationship between task displacement and changes in base wages, measured by the change at the 30th percentile, $\Delta \ln w_g(30th)$. The middle panel plots the relationship between task displacement and rent losses, measured by the additional wage decline above the 30th percentile, $\Delta \ln \bar{w}_g - \Delta \ln w_g(30th)$. The right panel repeats the exercise but purges wages of job-quality differences. All specifications include our baseline controls: gender and education dummies, sectoral demand and rent shifts, and exposure to manufacturing.

The right-most panel shows estimates of rent dissipation, as in the middle panel, but now also controlling for compensating differentials. Namely, the dependent variable $\Delta \ln \bar{w}_g - \Delta \ln w_g(30th)$ is constructed after purging wages of job quality. The point estimate of $\beta = -0.28$ (std. err. = 0.11) implies a rate of rent dissipation of 28%, which is similar to the estimate in the middle panel. This shows that accounting for compensating wage differentials does not affect our estimates of rent dissipation from automation.

Online Appendix Tables S2 and S3 confirm that these results are not sensitive to the use of the 30th percentile as our measure of base wages: we obtain similar estimates when using the 20th and 40th percentiles, or when we use the average wage decline between the 10th and 40th percentiles to estimate the change in base-wage changes due to automation. We also report similar point estimates for the rate of rent dissipation among noncollege and college workers, pointing to the pervasiveness of this phenomenon (and confirming that the loss of high-rent jobs is not confined to noncollege workers, even if the extent of this loss is significantly larger for noncollege groups).¹⁸

18. Another concern is that minimum wages or regulations may explain the weaker effects at the bottom of the within-group distribution in Figure V. Online Appendix S5 shows this is not the case: the U-shape is robust to controlling for minimum wage incidence or restricting the sample to groups with average wages

2. *Estimates of Rent Dissipation from Employment Shifts away from High-Rent Jobs.* In our theory, rent dissipation arises because automation displaces workers from high-rent tasks. Our second strategy uses proxies for rents to directly measure rent dissipation from these shifts in employment away from high-rent jobs.

We start with time-invariant measures of rents, denoted Rents_{gn} , which capture the extent of rents in job n (defined at the industry or industry $\times$ occupation level) for demographic group g . We then compute a second measure of rent dissipation as $\sum_n \text{Rents}_{gn} \Delta \ell_{gn}$, where $\Delta \ell_{gn}$ is the change in the share of hours worked by group g in job n between 1980 and 2016. A negative value for this measure indicates that workers from group g moved out of high-rent jobs and into low-rent jobs.

With these measures, we estimate a variant of [equation \(12\)](#):

$$(15) \quad \sum_n \text{Rents}_{gn} \Delta \ell_{gn} = \beta \text{Task displacement}_g + X_g \gamma + u_g.$$

The coefficient on task displacement captures the extent to which automation shifts exposed groups away from high-rent jobs, providing a direct estimate of rent dissipation and the associated decline in allocative efficiency.

Although this strategy requires proxies of job rents, it provides a more direct test of the core mechanism in our theory. In our model with fixed wedges, all rent dissipation comes from the inefficient displacement of workers from high-rent jobs, and this is exactly what this strategy measures. Comparing these estimates of rent dissipation to those from our first strategy thus serves as a check on our assumption that wedges are fixed and that rent dissipation operates through worker displacement. Under this hypothesis, this second strategy should yield the same magnitude of rent dissipation as the first.

We estimate [equation \(15\)](#) using four proxies for rents.

Our first proxy uses interindustry and occupation wage differentials in 1980. This approach builds on the interindustry wage-differentials literature, which documents that such differences are persistent and cannot be explained solely by worker characteristics or job amenities (see [Krueger and Summers 1988](#); [Katz and](#)

above \$10 in 1980. If anything, these strategies remove the imprecise negative effects at the bottom, suggesting that the dip at the 5th or 10th percentiles reflects wage floors compressing wages more in 1980 than in 2016.

Summers 1989). We measure the relative rent paid to workers from group g in industry i and occupation o as $\frac{\bar{w}_{gio}}{\bar{w}_g}$, where $\bar{w}_{gio}$ is the average wage of group g in industry i and occupation o in 1980.¹⁹

The top left panel of Figure VIII reports results using this first proxy. We estimate that a 10 percentage point increase in task displacement reduces group rents by 3.9%, reflecting a reallocation of workers away from higher-wage industries and occupations. The point estimate implies a rate of rent dissipation due to automation ($\frac{\mu_{Ag}}{\mu_g} - 1$ in the theory) of 39%—very close to the magnitude obtained with our first strategy, as hypothesized.

One concern with our first proxy is that part of the wage differentials could be due to differences in worker productivity across industry-occupation cells. Our second proxy addresses this issue by building on Gibbons and Katz (1992) and focusing on wage losses after job displacement. Because these compare the same worker before and after displacement, they are not affected by unobserved heterogeneity.²⁰ We compute wage losses for workers with at least one year of tenure before displacement who subsequently found a new job, using the CPS Displaced Worker Supplement. Given the small sample of displaced workers, we estimate wage losses by industry and six broad occupations, allowing losses to vary by gender and education (rather than at the more granular level of 500 demographic groups). We average these estimates over 1984–2022 to obtain our second proxy for Rents_{gn}.²¹

The top right panel of Figure VIII presents the results using this second proxy. A 10 percentage point increase in task displace-

19. This is equivalent to measuring worker rents by a group-specific industry $\times$ occupation intercept in a Mincer regression, while controlling for a full set of interactions between age, gender, education, race, and birthplace. Our baseline results use wage differentials for the 49 industries in our analysis and 300 detailed census occupations. Online Appendix Table S6 shows that the results are robust to using wage differentials computed for broader occupational groups or only across industries.

20. The estimates of rent dissipation using this proxy for rents also rule out other potential explanations for the U-shape in Figures V and VI. For example, an alternative explanation for the U-shaped pattern based on within-group heterogeneity in both absolute and comparative advantage could not explain why automation shifts workers out of high-rent jobs (with rents measured by wage losses from job displacement). This advantage is also shared by our estimates using quit rates to measure rents.

21. A higher tenure threshold reduces the sample of displaced workers, but does not affect our estimates. Online Appendix Table S6 establishes the robustness of our results to using varying levels of aggregation for this rent measure.

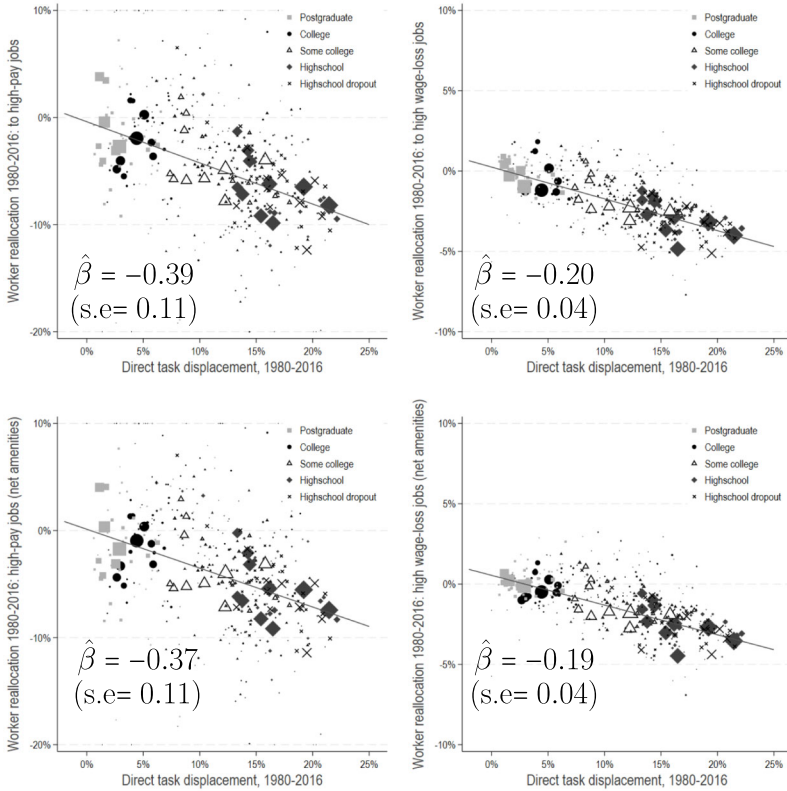


FIGURE VIII

Reduced-Form: Automation and Shifts away from High-Rent Jobs

The figure plots the relationship between automation and worker shifts away from high-rent jobs, measured as $\sum_n \text{Rents}_{gn} \Delta \ell_{gn}$. The top left panel uses industry and occupation wage differentials in 1980 as the proxy for rents. The top right panel uses wage losses after job displacement from the CPS Displaced Worker Supplement. The bottom panels repeat these analyses after purging job-quality differences from the rent proxies. All specifications include baseline controls: gender and education dummies, sectoral demand and rent shifts, and exposure to manufacturing.

ment is estimated to reduce group rents by 2%, as workers are re-allocated away from jobs with higher rents. This estimate implies a rent-dissipation rate from automation of 20%, once again close to the magnitudes obtained from our first strategy.

One drawback of the first two proxies is that they may capture compensating differentials. We address this concern in two

ways. First, building on our previous analysis, we purge both rent proxies of differences in job quality, measured using the job quality index from the 1977 Quality of Employment Survey. This index explains 9% of the variation in wage differentials and 14% of the variation in wage losses from displacement across industries. The bottom panels of [Figure VIII](#) report estimates using these residualized rent proxies. We obtain slightly smaller but very similar estimates of rent dissipation: 0.37 (std. err. = 0.11) using our first proxy and 0.19 (std. err. = 0.04) using our second proxy based on wage losses after displacement.

Our second approach is to use two additional proxies for rents that we expect to be unrelated to compensating differentials. This approach exploits differences in worker quit behavior, based on the idea that workers should be less likely to quit jobs that pay rents but not those that pay higher wages as compensating differentials. We compute two (time-invariant) proxies for quits from the CPS, as average employment-to-employment (EE) and voluntary employment-to-unemployment (EU) monthly quit rates by industry, occupation, gender, and education. The EE measure is averaged over 1994–2023 and the EU measure is averaged over 1976–2023.²² Consistent with our interpretation, quit rates are inversely associated with industry-occupation wage differentials (correlation of -0.72) and with wage losses after displacement (correlation of -0.54).

[Figure IX](#) reports estimates of a variant of [equation \(12\)](#) with $\text{Rents}_{gn} = -\text{Quit rate}_{gn}$, where the quit rate is computed either from the EE or the EU rates. In both cases, we see a clear negative relationship, suggesting that automation displaced workers from jobs with low quit rates. A 10 percentage point increase in task displacement pushes workers to jobs with a 0.15 percentage point higher EE rate and a 0.03 percentage point higher voluntary EU rate per month. This pattern suggests that automation is not shifting workers out of jobs with high compensating differentials but from jobs that they prefer to hold—hence jobs that used to pay higher rents.

22. Our data-cleaning procedure follows [Fujita, Moscarini, and Postel-Vinay \(2024\)](#). We compute average quit rates for the 49 industries and 300 detailed occupations. We also partial out time trends in the CPS from these industry $\times$ occupation averages. [Online Appendix Table S7](#) reports very similar results when we use average quit rates at different levels of aggregation.

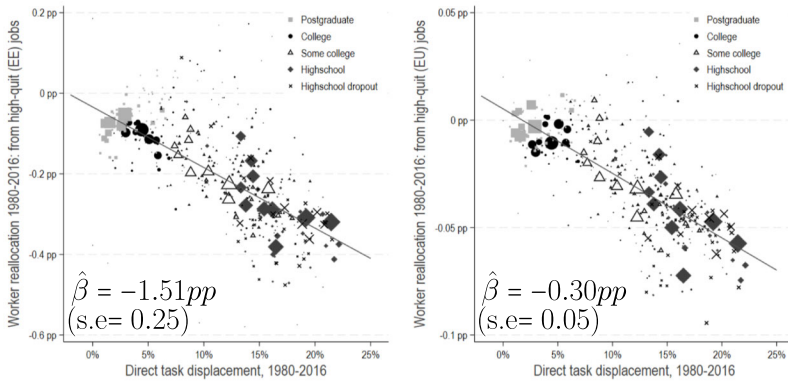


FIGURE IX

Reduced-Form: Automation and Shifts away from High-Rent Jobs (as Revealed by Quits)

The figure plots the relation between automation and worker shifts away from high-rent jobs, measured as $-\sum_n \text{Quit rate}_{gn} \Delta \ell_{gn}$. The left panel presents results using the employment-to-employment monthly transition rate as an inverse proxy for rents (computed from the Basic Monthly CPS). The right panel presents results using the voluntary employment-to-unemployment monthly transition rate as an inverse proxy for rents (computed from the Basic Monthly CPS). All specifications include baseline controls: gender and education dummies, sectoral demand and rent shifts, and exposure to manufacturing.

To facilitate the interpretation of these point estimates, we convert them into implied rents by using the empirical relationship between intra-industry wage differentials and quit rates. In our data, a 1 percentage point lower EE rate is associated with an 18.9% higher wage differential. Our estimates in the left panel of Figure IX then imply a rate of rent dissipation of $1.5 \times 18.9\% = 28.4\%$. Likewise, in our data, a 1 percentage point lower voluntary EU rate is associated with a 149% higher wage differential. Our estimates in the right panel of Figure IX then imply a rate of rent dissipation of $0.3 \times 149\% = 44.8\%$. These estimates are similar to those from our other two rent proxies and our first strategy.²³

23. We obtain similar results if we use estimates of the elasticity of separation rates with respect to wages from Bassier, Dube, and Naidu (2022) to do this conversion. Their paper estimates an elasticity of 2.1 of separation rates (into unemployment or other jobs) with respect to job-level wage differentials. Converted to a semi-elasticity, this implies that a 1 percentage point lower separation rate is associated with a 17% higher wage. Pooling the estimates from Figure IX for

The results from our second strategy confirm that automation caused significant rent dissipation, displacing exposed groups of workers from high-rent jobs.²⁴ Our estimates imply a rate of rent dissipation between 19% and 44.5%, with a central estimate of 35%.

3. Further Empirical Results and Checks. Our framework implies that automation is partially motivated by a desire to dissipate rents. Hence, all else equal, industries with higher rents should witness more automation. We provide evidence supporting this prediction in [Online Appendix Tables S12 and S13](#). Using our four proxies of rents, we show that high-rent industries exhibit larger declines in labor share and task displacement due to automation.²⁵

[Online Appendix S5](#) also provides additional results exploring the robustness of our findings. Besides providing tables summarizing all the estimates presented in figures here (see [Online Appendix Table S1](#)), we demonstrate the robustness of these estimates to different sets of covariates and also provide estimates using different definitions of routine jobs when measuring task displacement (see [Online Appendix Tables S8 and S9](#)). The [Online Appendix](#) also presents estimates for different subperiods (see [Online Appendix Tables S10 and S11](#)). We find some suggestive (but not conclusive) evidence in support of the view that rent dissipation was stronger early on in our sample period and has weakened over time. This pattern is consistent with a dynamic version of our theory. The same logic as in our model implies that highest-

EE and EU quits, we obtain a 1.8 percentage point higher overall quit rate, which implies a rate of rent dissipation of $1.8 \times 17\% = 30.6\%$.

24. All of our rent proxies point to a pervasive overall shift away from high-rent jobs for most demographic groups since 1980. The reduced-form evidence in this section shows that automation has been an important driver of this trend. For example, automation explains 33% (top left panel of [Figure VIII](#)) to 68% (left panel of [Figure IX](#)) of the extent to which different groups shifted away from high-rent jobs.

25. This evidence aligns with work by [Acemoglu and Restrepo \(2022a\)](#) and [Hémous et al. \(2025\)](#), which establish that higher wages (driven by demographic changes in the former paper and by labor market reforms in the latter paper) induce more automation. Our exercise additionally highlights the role of rents, which can induce inefficient automation. These results also align with [Doms, Dunne, and Troske \(1997\)](#), who show that establishments that pay higher wages subsequently adopt more automation technologies, though in practice there may be several different reasons other than worker rents that account for this association.

rent tasks should be automated first, and rent dissipation should be stronger early on and start declining as further advances in automation move on to displacing workers from medium-rent jobs.

IV. GENERAL EQUILIBRIUM EFFECTS OF AUTOMATION

This section explores the general equilibrium effects of automation in the U.S. labor market between 1980 and 2016 by estimating the ripple effects from automation and incorporating the sectoral reallocation of employment and output induced by automation. To do this, we first extend our model to a multi-sector economy and derive formulas describing the general equilibrium implications of automation in this context.

IV.A. Multi-Sector Economy

Consider an economy with multiple sectors $i \in \mathbb{I}$. Sector i produces output y_i with price p_i . Sectoral outputs are combined into a unique final good via an aggregator with a constant elasticity of substitution $\eta > 0$. Each of these goods is produced as in our single-sector economy, with good y_i produced by combining complementary tasks $x \in \mathcal{T}_i$,

$$y_i = \left(\frac{1}{M_i} \int_{\mathcal{T}_i} (M_i y_x)^{\frac{\lambda-1}{\lambda}} dx \right)^{\frac{\lambda}{\lambda-1}}.$$

The sets $\{\mathcal{T}_i\}_{i \in \mathbb{I}}$ are sector-specific, which is without loss since tasks can be relabeled.

The rest of the model is unchanged, and turns on how tasks in these industries are allocated to worker groups and capital. The details are provided in [Online Appendix S2](#). The definition of equilibrium is a direct generalization of the one in [Section II](#), with the only difference that sectoral prices are determined in equilibrium.

As before, new automation technologies are modeled as an exogenous increase in q_x from zero to q'_x for tasks in $\{\mathcal{A}_{gi}^T\}_{i \in \mathbb{I}, g \in \mathbb{G}}$, where $\mathcal{A}_{gi}^T$ denotes tasks that are feasible to automate (among those assigned to group g in industry i) and $\mathcal{A}_{gi}$ corresponds to the subset of tasks that are automated. We use δ_{gi} to denote the task displacement for group g in industry i and $\delta_g = \sum_i \frac{\ell_{gi}}{\ell_g} \delta_{gi}$ for the total task displacement across all industries, where ℓ_{gi} is group g 's employment in industry i . In addition, π_{gi} measures the cost savings from automating tasks in $\mathcal{A}_{gi}$, $\mu_{\mathcal{A}_{gi}}$ is the average rent in these tasks, and $\mu_{\mathcal{A}_g}$ is the average across industries.

The next proposition characterizes the general equilibrium effects of automation in the multi-sector economy.

PROPOSITION 6 (GENERAL EQUILIBRIUM EFFECTS OF AUTOMATION). Consider new automation technologies represented by (small interior) sets $\{\mathcal{A}_{gi}^T\}_{i \in \mathbb{I}, g \in \mathbb{G}}$. Let $\{\mathcal{A}_{gi}\}_{i \in \mathbb{I}, g \in \mathbb{G}}$ be given by [equation \(4\)](#) and the corresponding direct effects of automation be given by $\{\delta_{gi}\}_{i \in \mathbb{I}, g \in \mathbb{G}}$, $\{\pi_{gi}\}_{i \in \mathbb{I}, g \in \mathbb{G}}$, $\{\mu_{\mathcal{A}_{gi}}\}_{i \in \mathbb{I}, g \in \mathbb{G}}$. The first-order impact on base wages w_g , sectoral prices p_i , and output y are given by the solution to the system of equations:

$$(16) \quad d \ln w_g = \Theta_g \text{stack} \left(\frac{1}{\lambda} d \ln y - \frac{1}{\lambda} \delta_j + \frac{1}{\lambda} \sum_i \frac{\ell_{ji}}{\ell_j} (\lambda - \eta) d \ln p_i \right) \quad \text{for } g \in \mathbb{G}$$

$$(17) \quad d \ln p_i = \sum_g s_{gi} d \ln w_g - \sum_g s_{gi} \frac{\mu_{\mathcal{A}_{gi}}}{\mu_{gi}} \delta_{gi} \pi_{gi} \quad \text{for } i \in \mathbb{I}$$

$$(18) \quad \sum_g s_g d \ln w_g = \sum_i s_{yi} \sum_g s_{gi} \frac{\mu_{\mathcal{A}_{gi}}}{\mu_{gi}} \delta_{gi} \pi_{gi},$$

where s_{gi} is the share of group g 's earnings in industry i 's output and s_{yi} is the output share of industry i in GDP. Moreover, the change in group rents is given by

$$(19) \quad d \ln \mu_g = \mathcal{M}_g \text{stack} \left(\frac{1}{\lambda} d \ln y - \frac{1}{\lambda} \delta_j + \frac{1}{\lambda} \sum_i \frac{\ell_{ji}}{\ell_j} (\lambda - \eta) d \ln p_i \right) - \left(\frac{\mu_{\mathcal{A}_g}}{\mu_g} - 1 \right) \delta_g + \sum_i \left(\frac{\mu_{gi}}{\mu_g} - 1 \right) \frac{\ell_{gi}}{\ell_g} (\lambda - \eta) d \ln p_i \quad \text{for } g \in \mathbb{G},$$

and the change in aggregate consumption is $d \ln c = \frac{1}{s_\ell} d \ln \text{TFP}$, where s_ℓ is the economy-wide labor share and $d \ln \text{TFP}$ is the contribution of automation to TFP, given by

$$(20) \quad d \ln \text{TFP} = \sum_i s_{yi} \sum_g s_{gi} \frac{\mu_{\mathcal{A}_{gi}}}{\mu_{gi}} \delta_{gi} \pi_{gi} + \sum_g s_g d \ln \mu_g.$$

From this proposition, the general equilibrium effects of automation on group wages can be summarized as:

$$\begin{aligned}
 d \ln \bar{w}_g = & (\Theta_g + \mathcal{M}_g) \text{ stack} \left(\frac{1}{\lambda} d \ln y - \frac{1}{\lambda} \delta_j \right. \\
 & \left. + \frac{1}{\lambda} \sum_i \frac{\ell_{ji}}{\ell_j} (\lambda - \eta) d \ln p_i \right) \\
 (21) \quad & - \left(\frac{\mu_{\mathcal{A}_g}}{\mu_g} - 1 \right) \delta_g + \sum_i \left(\frac{\mu_{gi}}{\mu_g} - 1 \right) \frac{\ell_{gi}}{\ell_g} (\lambda - \eta) d \ln p_i.
 \end{aligned}$$

Relative to [Proposition 5](#), these formulas account for the impact of automation on sectoral prices and the resulting changes in sectoral composition of the economy (for example, automation in one industry can now reallocate expenditure away from that industry). [Equation \(17\)](#) shows that changes in sectoral prices depend on base-wage changes (from Shephard's lemma) and cost savings from automation. Induced changes in sectoral composition affect wages because sectors differ in terms of their labor demands. This is measured by the term $\sum_i \frac{\ell_{ji}}{\ell_j} (\lambda - \eta) d \ln p_i$ in [equations \(16\)](#) and [\(21\)](#). The sectoral reallocation also influences the distribution of rents, as captured by the term $\sum_i \left(\frac{\mu_{gi}}{\mu_g} - 1 \right) \frac{\ell_{gi}}{\ell_g} (\lambda - \eta) d \ln p_i$ in [equations \(19\)](#) and [\(21\)](#).

IV.B. Measurement and Estimation

The formulas in [Proposition 6](#) allow us to compute the first-order effects of automation in terms of the following objects:

- (i) the task displacement of automation across industries δ_{gi} and on aggregate δ_g ;
- (ii) the rent dissipation due to automation, $\frac{\mu_{\mathcal{A}_{gi}}}{\mu_g}$ and $\frac{\mu_{\mathcal{A}_g}}{\mu_g}$;
- (iii) the cost savings from automation, π_{gi} ;
- (iv) the propagation matrix, Θ , and the rent-impact matrix, $\mathcal{M}$.

In addition, the formulas involve initial factor shares, which we compute from the BEA and census data, and the elasticities of substitution between tasks, λ , and sectors, η , which we set as $\lambda = 0.5$ from [Humlum \(2020\)](#) and $\eta = 0.2$ from [Buera, Kaboski, and Rogerson \(2015\)](#).

The rest of our approach requires three simplifying choices. First, we impose [Assumption 3](#). Second, we assume that cost sav-

ings from the adoption of industrial robots in manufacturing—about 30% in [Acemoglu and Restrepo \(2020\)](#)—applies across the board and set $\pi_{gi} = 30\%$ for all industries and groups.²⁶ Third, we assume the rate of rent dissipation $\frac{\mu_{\delta gi}}{\mu_{gi}}$ is the same across industries and groups. In particular, we set $\frac{\mu_{\delta gi}}{\mu_{gi}} = 1 + \rho$ and treat ρ —the common rate of rent dissipation—as a parameter to be estimated.

Under these assumptions, aggregate task displacement δ_g is again given by [equation \(13\)](#), as in our reduced-form analysis. The only difference now is that in the reduced form we imposed the value of ρ , whereas here we estimate it directly.²⁷

1. *Estimating ρ , Θ , $\mathcal{M}$.* Our model delivers two equations in terms of changes in base wages and rents as a function of task displacement and other equilibrium objects (such as other groups' wage changes):

$$(22) \quad \Delta \ln w_g = \beta_0 - \frac{1}{\lambda} \delta_g + \beta X_g + \frac{1}{\lambda} \mathcal{J}_{\Gamma, g} \text{stack}(\Delta \ln w_j) + u_g$$

$$(23) \quad \Delta \ln \mu_g = \beta_0^\mu - \rho \delta_g + \beta^\mu X_g^\mu + \mathcal{J}_{\mu, g} \text{stack}(\Delta \ln w_j) + e_g.$$

[Equation \(22\)](#) describes the change in base wages in terms of common shifts β_0 (such as the expansion of GDP, $\Delta \ln y$), the task displacement from automation δ_g , other shifts in technology X_g (such as sectoral productivity shifts), and ripple effects. These are captured by $\mathcal{J}_{\Gamma, g} \text{stack}(\Delta \ln w_j)$, where $\mathcal{J}_{\Gamma}$ is the Jacobian summarizing how task shares adjust in response to wage changes, and $\mathcal{J}_{\Gamma, g}$ is its g th row. The error term u_g represents other changes in technology or labor supply.

[Equation \(23\)](#) describes changes in group rents in terms of a constant β_0^μ , rent dissipation from automation, $\rho \delta_g$, the direct impact on rents of other shifts in technology, X_g^μ , and ripple effects,

26. We do this because we lack disaggregated data on the costs of and productivity gains from other automation technologies. This feature can be improved in future work leveraging industry-specific estimates.

27. Consistent with this, the industry-level task displacement for group g in industry i , which is needed to compute change in sectoral prices in [Proposition 6](#), is measured as (see [Online Appendix S5.7](#) for the derivation):

$$\delta_{gi} = \text{RCA}_{gi}^{\text{routine}} \frac{-d \ln s_{\ell i}^A}{1 + s_{\ell i} (\lambda - 1) \pi} \frac{1}{1 + \rho}.$$

$\mathcal{J}_{\mu,g}$ $\text{stack}(\Delta \ln w_j)$, now with $\mathcal{J}_{\mu}$ representing the rent-impact Jacobian. The error term e_g corresponds to omitted factors affecting rents (for example, due to changes in wedges coming from other sources).

In essence, these equations define a factor demand system, describing how the base wage and average rent of a group are affected by wage changes of other groups and demand shifters, including task displacement and sectoral composition terms. The key unknowns in these equations can be written in the form of the propagation and rent-impact matrices: $\Theta = (\mathbb{1} - \frac{1}{\lambda} \mathcal{J}_{\Gamma})^{-1}$ and $\mathcal{M} = \mathcal{J}_{\mu} (\mathbb{1} - \frac{1}{\lambda} \mathcal{J}_{\Gamma})^{-1}$. These matrices, which fully summarize the ripple effects, can be estimated under similar assumptions to those we made in our reduced-form analysis. However, without further restrictions, each one of the matrices Θ and $\mathcal{M}$ include 500×500 entries—since the effects of task displacement experienced by g on group g' could be very different than its effect on group g'' .

To make progress, we put additional structure and parameterize the entries of the Jacobian matrices in terms of job and demographic group-level similarities. For the diagonal terms, we suppose that

$$\begin{aligned} \mathcal{J}_{\Gamma,g,g} = (s_g - 1) \varphi - \sum_n \sum_{g' \neq g} \frac{\ell_{gn}}{\ell_g} s_{g'n} [\theta + \theta_{\text{job}} \text{job similarity}_{gg'} \\ + \theta_{\text{edu-age}} \text{edu-age similarity}_{gg'}] \end{aligned}$$

and the off-diagonal terms, for $g' \neq g$, are assumed to take the form:

$$\begin{aligned} \mathcal{J}_{\Gamma,g,g'} = s_{g'} \varphi + \sum_n \frac{\ell_{gn}}{\ell_g} s_{g'n} [\theta + \theta_{\text{job}} \text{job similarity}_{gg'} \\ + \theta_{\text{edu-age}} \text{edu-age similarity}_{gg'}]. \end{aligned}$$

This parameterization imposes that competition between groups for marginal tasks takes place within job categories, defined in the data at the level of 16 industries and six occupations (denoted by n). The effects of competition from demographic group g' on group g in job category n are then assumed to depend on the importance of this job category for group g , $\frac{\ell_{gn}}{\ell_g}$, and on the extent of group g' 's impact on this job category, represented by the share of earnings in job category n accruing to group g' , $s_{g'n}$. (We measure both of these from the 1980 census.) This captures the intuitive notion

that groups with greater shares should generate more competitive pressure on other groups in the same job category.

The parameters θ , θ_{job} , $\theta_{\text{edu-age}} \geq 0$ summarize the extent of competition for marginal tasks across groups. Specifically, θ represents competition for tasks common to all workers in a job category. For example, if worker productivity for all tasks in job category n were drawn from independent Fréchet distributions with shape parameter ν , then we would have $\theta = \nu + 1 - \lambda$. The parameters θ_{job} and $\theta_{\text{edu-age}}$ then allow competition to be more intense among groups that hold similar jobs elsewhere in the economy (suggesting a greater overlap in their skills) or that are of a similar age and education (as in [Card and Lemieux 2001](#)).²⁸

Note also that in these specifications, the constant $\varphi \geq 0$ controls the extent of competition between capital and workers for marginal tasks. In this parameterization, the aggregate elasticity of substitution between capital and labor is $\sigma = \lambda + \varphi$ (capturing the sum of the elasticities of substitution between tasks and between demographic groups in marginal tasks). We set $\varphi = 0.1$ externally to match estimates of $\sigma = 0.6$ from [Oberfeld and Raval \(2021\)](#).

For the rent Jacobian, we follow a similar strategy and parameterize it as

$$\mathcal{J}_{\mu, g, g} = - \sum_n \sum_{g' \neq g} \left(\frac{\bar{w}_{gn}}{\bar{w}_g} - 1 \right) \frac{\ell_{gn}}{\ell_g} s_{ng'} [\theta + \theta_{\text{job}} \text{ job similarity}_{gg'} + \theta_{\text{edu-age}} \text{ edu-age similarity}_{gg'}],$$

and the off-diagonal terms, for $g' \neq g$, as

$$\mathcal{J}_{\mu, g, g'} = \sum_n \left(\frac{\bar{w}_{gn}}{\bar{w}_g} - 1 \right) \frac{\ell_{gn}}{\ell_g} s_{ng'} [\theta + \theta_{\text{job}} \text{ job similarity}_{gg'} + \theta_{\text{edu-age}} \text{ edu-age similarity}_{gg'}].$$

The term $\left(\frac{\bar{w}_{gn}}{\bar{w}_g} - 1 \right)$ proxies for rents earned by group g in job category n and accounts for whether competition from demographic group g' takes place at jobs where g workers earn above average rents. We compute this from observed wages in the 1980 census

28. We measure job similarity_{gg'} by the cosine similarity in jobs performed by groups g' and g in the 1980 census (over industry-occupation cells). For edu-age similarity_{gg'}, we estimate a Mincer wage regression and compute the similarity between g and g' as one minus the absolute difference in their predicted wages—based on each group's education and experience—rescaled to lie in $[0, 1]$.

as well. The rows of the rent Jacobian sum to zero, which imposes the restriction that substitution of capital for labor at marginal tasks does not affect group rents on average.

With these parametric forms, the system of [equations \(22\) and \(23\)](#) enable us to estimate ρ , θ , θ_{job} , and $\theta_{\text{edu-age}}$ using generalized method of moments (GMM). As in our reduced-form analysis, we measure the change in base wages $\Delta \ln w_g$ by the change at the 30th percentile of the within-group distribution and take the decline in wages above the 30th percentile of exposed groups as a measure for the change in average rents, $\Delta \ln \mu_g$. Formally, the orthogonality conditions for the GMM estimation are essentially the same as in our reduced-form analysis:

$$\delta_g, X_g, X_g^\mu \perp u_j, e_j \text{ for all } g,$$

except that we are now simultaneously estimating the ripple effects.²⁹ Once these parameters are estimated, the propagation and rent-impact matrices can be recovered from their definitions:

$$\Theta = \left(\mathbb{I} - \frac{1}{\lambda} \mathcal{J}_\Gamma \right)^{-1} \text{ and } \mathcal{M} = \mathcal{J}_\mu \left(\mathbb{I} - \frac{1}{\lambda} \mathcal{J}_\Gamma \right)^{-1}.$$

[Online Appendix Table S14](#) reports our GMM estimates for ρ , θ , θ_{job} , $\theta_{\text{edu-age}}$. Here we summarize the main findings:

(i) The common rent dissipation coefficient is estimated as $\hat{\rho} = 0.35$ (std. err. = 0.12). This aligns with our reduced-form estimates in [Section III](#).

(ii) The propagation matrix has an average diagonal of 0.34, and the row sum of the off-diagonal terms is 0.6. This implies that exposed workers bear 40% of the incidence of automation, with the rest shifted to others via ripple effects.

(iii) The entries of the rent-impact matrix are small, which implies that ripple effects on group rents are small.

IV.C. General Equilibrium Effects of Automation

This section reports our estimates of the effects of automation on wages, rents, and TFP in the United States between 1980 and 2016. The calculations use the formulas from [Proposition 6](#), our measures of task displacement, and the parameter estimates from the previous subsection. [Table I](#), column (1) summarizes the data, and column (2) presents our estimates.

29. Recall that [equation \(22\)](#) is in essence a factor demand system, which can be estimated using various demand shifters. In the trade literature, these shifters come from changes in trade costs, for example induced by the evolution of tariffs (see [Adao, Costinot, and Donaldson 2017](#)). Our approach instead uses different groups' uneven exposure to automation but operates with the same logic.

TABLE I
ESTIMATED EFFECTS OF NEW AUTOMATION TECHNOLOGY, 1980–2016.

	Data 1980–2016 (1)	Baseline estimates (2)	Low rent dissipation (20%) (3)	High rent dissipation (50%) (4)	Imposing aggregate elasticity of substitution of 1.6 between groups (5)	Higher elasticity of substitution between industries ($\eta = 0.75$) (6)
Panel A: Wage structure						
Change in average wages, $d \ln \bar{w}_g$ (%)	29.15	0.456	1.91	−0.705	0.497	0.61
Change in base wages, $d \ln w_g$ (%)		4.54	4.54	4.54	4.54	4.54
Change in group rents, $d \ln \mu_g$ (%)		−4.08	−2.63	−5.24	−4.04	−3.92
Wage changes for noncollege men (%)	−6.5	−8.0	−6.8	−9.0	−6.9	−6.5
Base wage changes (%)		−2.4	−3.2	−1.7	−1.3	−0.9
Rent dissipation (%)		−5.6	−3.6	−7.3	−5.6	−5.4
Wage changes for noncollege women (%)	10.6	−2.2	−0.8	−3.3	−1.5	−2.6
Base wage changes (%)		2.4	2.1	2.6	3.0	1.9
Rent dissipation (%)		−4.6	−2.9	−5.9	−4.6	−4.5
Between-group wage changes explained (%)						
Due to changes in industry composition (%)		6.72	7.30	6.26	5.92	−4.76
Adding direct displacement effects (%)		63.49	71.16	57.35	62.68	52.00
Accounting for ripple effects (%)		41.79	46.84	37.75	35.08	34.23
Accounting for rent dissipation (%)		51.87	53.38	50.66	45.39	44.13

TABLE I
CONTINUED

	Data 1980–2016 (1)	Baseline estimates (2)	Low rent dissipation (20%) (3)	High rent dissipation (50%) (4)	Imposing aggregate elasticity of substitution of 1.6 between groups (5)	Higher elasticity of substitution between industries ($\eta = 0.75$) (6)
Panel B: Aggregates						
Change in GDP per capita, $d \ln y$ (%)	70	14.41	15.90	13.22	14.50	14.02
Change in TFP, $d \ln TFP$ (%)	35	0.304	1.27	-0.47	0.331	0.41
Cost-saving gains (%)		3.02	3.02	3.02	3.02	3.02
Changes in allocative efficiency (%)		-2.72	-1.75	-3.49	-2.69	-2.61
Inefficient rent dissipation (%)		-2.70	-1.73	-3.47	-2.7	-2.7
Change in consumption, $d \ln c$ (%)	70	0.456	1.91	-0.705	0.497	0.61
Change in $\frac{k}{y}$ ratio (%)	30.0	27.9	28.0	27.8	28.0	26.8

Notes. This table summarizes the estimated effects of automation on group wages and aggregates from 1980 to 2016. The estimates are computed using the formulas in Proposition 6. Column (1) reports data for the United States for comparison. The wage data is from the U.S. Census and the ACS. The data on GDP and capital are from the BLS. The data on consumption is from the BEA and the data for TFP is from Fernald (2014). Column (2) provides our baseline estimates. Column 3 assumes a lower rate of rent dissipation of 20%. Column (4) (4) assumes a higher rate of rent dissipation of 50%. Column (5) assumes $\theta = 1.6 - \lambda$ and sets $\theta_{job} = \theta_{dis-age} = 0$, imposing an aggregate elasticity of substitution between worker groups of 1.6. Column (6) uses the baseline specification from column (2) but sets the elasticity of substitution between industries η to 0.75.

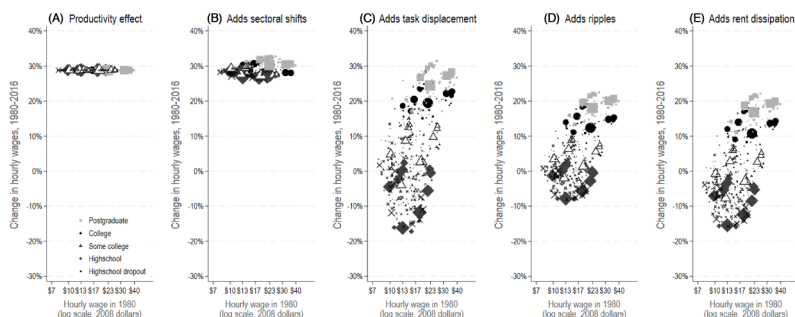


FIGURE X

Wage Effects of Automation

The figure shows estimates of automation's impact on between-group wage changes through different channels, beginning with the productivity effect in Panel A and adding components cumulatively until Panel E. In all panels, the horizontal axis sorts groups by their hourly wage in 1980.

1. Effects on Wages and Group Rents. Table I, Panel A reports our estimates for wages, which are also depicted in Figure X. Using equation (21), the figure decomposes the change in average wages for the 500 demographic groups into the different channels through which automation operates. The vertical axis shows wage changes over 1980–2016, and the horizontal axis orders groups by their average hourly wages in 1980.

Figure X, Panel A shows only the productivity effect of automation, given by $\frac{1}{\lambda} d \ln y$. We estimate that aggregate output increased by 14.4% over 1980–2016 due to automation, and this raised wages by 28.8% for all demographic groups. We estimate that automation increased aggregate output by 14.4% between 1980 and 2016, raising wages by 28.8% for all demographic groups. Panel B adds the effects of sectoral shifts induced by automation, computed as $\frac{1}{\lambda} d \ln y + \frac{1}{\lambda} \sum_i \frac{\ell_{gi}}{\ell_g} (\lambda - \eta) d \ln p_i$. The modest dispersion of wages across groups in this panel indicates that the overall contribution of automation-driven sectoral shifts to wage inequality was limited.

Panel C adds the direct displacement effects from automation, thus plotting $\frac{1}{\lambda} d \ln y + \frac{1}{\lambda} \sum_i \frac{\ell_{gi}}{\ell_g} (\lambda - \eta) d \ln p_i - \frac{1}{\lambda} \delta_g$. Consistent with the reduced-form evidence, task displacement has significant effects on the wage structure and generates much greater dispersion compared to Panels A and B. Although groups less ex-

posed to automation experience real wage gains, we see large declines in the real wages of highly exposed groups.

Panel D illustrates the role of ripple effects, by plotting Θ_g stack $(\frac{1}{\lambda} d \ln y + \frac{1}{\lambda} \sum_i \frac{\ell_{ji}}{\ell_j} (\lambda - \eta) d \ln p_i - \frac{1}{\lambda} \delta_j)$. Ripple effects play an equalizing role, as can be seen from the fact that the dispersion in wage changes across groups is less pronounced than in Panel C. This is because automation lowers the relative wages of displaced groups, triggering a reassignment of marginal tasks toward them and away from others. Ripple effects are sizable and spread about two-thirds of the incidence of automation across groups.

Panel E completes the picture by adding changes in group rents, $d \ln \mu_g$, so we can now see the full effects of automation. The figure shows sizable rent dissipation, which reduces wages among highly exposed groups by about 10%.³⁰ In contrast to ripple effects, rent dissipation plays an unequalizing role. This is because, as explained in the theory, exposed groups bear the full incidence of rent dissipation.

To further illustrate the contribution of rent dissipation, [Figure XI](#) plots actual group-level wage changes against our estimates of wage changes due to automation between 1980 and 2016. The left panel shows the effects of automation working only through base wages, omitting rent dissipation. The right panel depicts the full effect, including rent dissipation. Changes in base wages alone explain 42% of observed wage changes, while estimates incorporating rent dissipation explain a total of 52%. These results imply that automation accounts for 52% of the rise in between-group wage inequality over this period, with about one-fifth of the effect being due to rent dissipation.³¹

The effects of rent dissipation are especially important for explaining the lack of real wage growth among low-education groups. On the left panel, several demographic groups lie below the 45-degree line, but far fewer do so on the right. Without

30. Most of the decline in rents across groups reflects direct rent dissipation—automation targeting jobs with above-average rents—rather than indirect effects working through the rent-impact matrix.

31. The 42% number is lower than the 50% estimate in [Acemoglu and Restrepo \(2022b\)](#), which used a model with fully competitive labor markets. If data are generated with imperfect labor markets but we impose a fully competitive model to interpret it, rent dissipation effects can load on the direct displacement term, exaggerating its role.

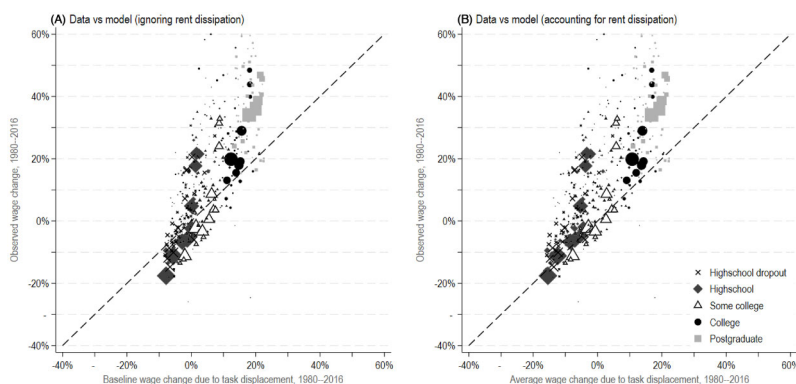


FIGURE XI

Wage Effects of Automation: Model Versus Data

The left panel compares predicted wage changes, excluding rent dissipation, to observed changes from 1980–2016. The right one includes rent dissipation and shows the full model-predicted wage changes against data.

accounting for rent dissipation, automation cannot explain the real wage declines experienced by many low-education groups between 1980 and 2016. Once rent dissipation is included, our estimates account for most of the observed real wage stagnation and decline over this period.

As a further illustration, Table I reports our estimates of the change in wages and rents for noncollege men and women. Without rent dissipation, automation would have led to a 2.4% decline in the real wage of noncollege men during 1980–2016 (compared with 6.5% in the data) and a 2.4% increase in the real wage of women. Once rent dissipation is factored in, we estimate an 8% decline for noncollege men and a 2.2% decline for noncollege women.

Rent dissipation also implies that automation generated virtually no aggregate wage gains during this period. Absent rent dissipation, automation would have raised the average real wage of U.S. workers by about 4.5% between 1980 and 2016. Once rent dissipation is taken into account, the predicted gain falls to 0.5%.

2. *Effects on Productivity and Consumption.* Table I, Panel B reports our estimates for aggregate output, consumption, and TFP. Our benchmark value for cost savings of $\pi = 30\%$, combined with the extent of automation estimated in the data, implies that

automation generated a 3% increase in TFP via cost savings. However, our estimate for $\frac{\mu_{\text{adg}}}{\mu_g}$ of 35% also means that this automation created inefficient rent dissipation. Overall, our estimates imply that automation worsened allocative efficiency by 2.7% during this period. This inefficiency offset 90% of potential cost savings and led to a net contribution of automation to TFP of only 0.3% between 1980 and 2016.³²

Turning to consumption, and using the formula $d \ln c = \frac{1}{s_g} d \ln \text{TFP}$, we estimate that automation increased aggregate consumption by about 0.46% during 1980–2016. This is smaller than the estimated increase in GDP of 14.4%, because GDP growth also includes the effects of greater investment required for producing capital equipment. Indeed, according to our estimates, automation raised the capital-output ratio by 27.9%. The increase in investment is in line with BLS data, which indicate a 30% increase in the aggregate capital stock relative to GDP.

3. Robustness. Table I reports a number of robustness checks. Column (3) depicts estimates obtained by assuming a rate of rent dissipation of $\frac{\mu_{\text{adg}}}{\mu_g} = 20\%$, as opposed to our baseline estimate of 35%. The impact of rent dissipation on wages and TFP is less pronounced in this case, but it still offsets 60% of the cost savings and average wage gains from new automation between 1980 and 2016. Column (4) goes in the other direction and reports the consequences of a rate of rent dissipation of $\frac{\mu_{\text{adg}}}{\mu_g} = 50\%$. At this rate, we estimate that automation would have reduced average wages by 0.7%, the wages of noncollege men by 9%, and the wages of noncollege women by 3.3%. We also estimate that the worsening allocative efficiency due to rent dissipation would have overwhelmed the cost-saving gains from automation, reducing aggregate TFP by 0.47% during 1980–2016.

Column (5) reports estimates from a specification of the Jacobians setting $\theta = 1.6 - \lambda$, $\theta_{\text{job}} = \theta_{\text{edu-age}} = 0$, which imposes a common elasticity of substitution between groups of 1.6, as in Katz

32. In the data, U.S. TFP increased by 30% over this time period. Our estimates therefore imply that almost all of this increase was due to other technologies—not due to automation. These other technologies include those that create new tasks for workers or new products for consumers, those that provide better information or tools to workers, raising their productivity in tasks already assigned to them, or improvements in capital productivity in tasks that were already automated (see Acemoglu and Restrepo 2018).

and Murphy (1992). We continue to assume the baseline rate of rent dissipation of 35%. The effects of automation on wages and productivity are similar to the baseline.

Column (6) reports estimates obtained by using a larger elasticity of substitution between sectoral outputs, $\eta = 0.75$. This parameter does not affect our conclusions, in part because sectoral shifts are not a major channel via which automation affects the wage structure, as shown in Figure X.

V. CONCLUSION

This article developed a framework in which labor market rents shape the consequences of automation. Our main insight is that firms tend to automate high-rent tasks first. Consequently, compared with a competitive benchmark, automation amplifies wage declines for exposed groups, reduces within-group wage dispersion, and can worsen allocative efficiency, offsetting much of the direct cost savings from automation and potentially lowering TFP and welfare.

We documented that in the U.S. data between 1980 and 2016, there is a robust pattern indicating the predicted U-shaped within-group wage response and a reallocation of workers away from high-rent jobs. A quantitative exercise exploiting the structure of our theoretical framework implies that automation overall explains 52% of the rise in between-group inequality since 1980, but only 42 percentage points of this is driven by competitive forces, with the remaining 10 percentage points accounted for by rent dissipation. The efficiency losses from rent dissipation offset nearly all of the cost savings from automation, leaving close to zero gains in TFP, average wages, and consumption due to automation since the 1980s.

Our analysis centers on the interplay between automation and labor market rents. There are several interesting areas for future work, including (i) how automation interacts with other distortions, such as monopsony, product market power, financial frictions, and investment taxes/subsidies (see Acemoglu, Manera, and Restrepo 2020); (ii) how other shocks, such as outsourcing, international trade, and immigration, reshape the wage structure—not only through competitive labor demand channels but also via rent dissipation, which can be analyzed using our theoretical framework; (iii) how matched employer-employee data, together with information on technology adoption, balance sheets, and product

and labor market power of firms, can be leveraged to test additional implications of automation in the presence of labor market distortions; and (iv) how labor market institutions and imperfections shape firms' incentives to adopt different types of technologies, beyond automation, including those introducing new tasks and collaborative tools for workers.

SUPPLEMENTARY MATERIAL

An Online Appendix for this article can be found at *The Quarterly Journal of Economics* online.

DATA AVAILABILITY

The data and code underlying this article is available in Acemoglu and Restrepo (2026), in the Harvard Dataverse, <https://doi.org/10.7910/DVN/SNK8T6>.

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